

Growth in early infancy drives optimal brain functional connectivity which predicts cognitive flexibility in later childhood

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This **important** study details changes in the brain functional connectivity in a longitudinal cohort of Gambian children assessed outside a lab setup with functional near-infrared spectroscopy (fNIRS) from age 5 to 24 months, in relation to early physical growth and cognitive flexibility capacities at preschool age. While evidence supporting conclusions on the evolution of brain connectivity are **solid**, the statistical power was insufficient to perform proper analyses of longitudinal data and link the connectivity trajectories with early adverse conditions such as undernutrition and later cognitive development. This study will be of significant interest to neuroscientists, psychologists and neuroimaging researchers working on infant development in relation to environmental factors.

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Abstract

Functional brain network organization, measured by functional connectivity (FC), reflects key neurodevelopmental processes for healthy development. Early exposure to adversity, e.g. undernutrition, affects neurodevelopment, observable via disrupted FC, and leads to poorer outcomes from preschool age onward. We assessed longitudinally the impact of early growth trajectories on developmental FC in a rural Gambian population from age 5 to 24 months. To investigate how these early trajectories relate to later childhood outcomes, we assessed cognitive flexibility at 3-5 years. We observed that early physical growth before the fifth

month of life drove optimal developmental trajectories of FC that in turn predicted cognitive flexibility at pre-school age. In contrast to previously studied developmental populations, this Gambian sample exhibited long-range interhemispheric FC that decreased with age. Our results highlight the measurable effects that poor growth in early infancy has on brain development and the subsequent impact on pre-school age cognitive development, underscoring the need for early life interventions throughout global settings of adversity.

Introduction

The first 1000 days of life are of paramount importance for human brain (1) and body development (2, 3). Early adversity negatively impacts infant development, which leads to long-term consequences from delayed milestones in childhood development to lowered productivity at the individual and societal levels (4–6). Therefore, assessing trajectories of brain development in infants exposed to early adversity is essential to understand and mitigate effects on brain development and subsequent outcomes at school age and beyond (7, 8).

Early undernutrition engenders detrimental effects on a range of cognitive skills (9), with long-lasting effects through adulthood (10). Interventions using therapeutic feeding regimens and supplements have been shown to have a strong causal impact on childhood undernutrition, brain development, and subsequent neurodevelopmental outcomes (11–13). Despite numerous attempts of interventions, global rates of undernutrition remain high (14), with infants in low- and middle-income countries (LMICs) at greatest risk (15). Up to one-third of infants in LMICs are at risk of not meeting standard milestones of social, motor, linguistic, and cognitive development by pre-school age (16, 17), which can lead to lifelong consequences that impact social and economic development at individual, national, and global scales (18). Thus, it is essential to study neurodevelopment throughout the first two years in regions with high rates of undernutrition in order to optimize early interventions. To identify effective early interventions, we must first identify the mechanisms that link undernutrition to later cognitive outcomes. Effects of undernutrition on the developing brain may manifest through alterations in function brain networks (19), as shown in school-age children exposed to chronic poverty (20). However, our understanding of the relationships between early undernutrition, brain connectivity, and later cognitive outcomes is still very limited.

Functional brain network organization is commonly assessed with resting-state functional connectivity (FC) via temporal correlations in brain activity as measured with functional magnetic resonance imaging (fMRI) (21, 22). Based on data from fMRI, current models hypothesize that FC patterns mature throughout early development (23–27), where in typically developing brains, adult-like networks emerge over the first years of life as long-range functional connections between pre-frontal, parietal, temporal, and occipital regions become stronger and more selective (28–31). This maturation in FC has been shown to be related to the cascading maturation of myelination and synaptogenesis (32, 33) - fundamental processes for healthy brain development (34). Therefore, disrupted patterns of FC may reflect disrupted anatomical maturation of brain circuits and systems. For example, preterm infants with severe diffuse white matter impairment exhibited lower levels of FC in executive function networks compared with preterm infants with no or mild white matter impairment (35). Importantly, normative developmental patterns may be disrupted and even reversed in clinical conditions that impact development; e.g., increased short-range and reduced long-range FC have been observed in preterm infants (36) and in children with autism spectrum disorder (37, 38).

While widely used neuroimaging modalities such as fMRI can elucidate typical and atypical developmental trajectories, they are often poorly suited for lower resource neuroimaging contexts due to limited portability, need for specialized facilities, and the high cost associated with these methods (28, 36, 39). To address the need for neuroimaging studies in lower resource

environments including (but not limited to) in LMICs, researchers have increasingly turned to more portable tools, including electroencephalography (EEG) (40), functional near-infrared spectroscopy (fNIRS) (41–43) and diffuse correlation spectroscopy (DCS) (44). These tools enable the creation of mobile neuroimaging laboratories that can be deployed virtually anywhere, eliminating practical constraints imposed by costly and immobile methods. Indeed, research based in The Gambia, Guinea-Bissau, Bangladesh and Ivory Coast have established feasibility of these tools to perform brain imaging studies outside of a specialized lab environment, with recent studies reporting altered cortical physiology related to early adversity and undernutrition (40, 43–45). For example, using EEG, Xie and colleagues showed that in 2-3 year old Bangladeshi children, growth faltering was associated with FC in the theta and beta frequency bands, which was negatively related to children's IQ score at 4 years (46). With regards to fNIRS, it has a better spatial resolution and anatomical specificity than EEG, thus providing more precise and reliable localization of brain networks (47, 48). Moreover, fNIRS facilitates testing in awake and engaged infants, making the results more comparable with adult FC studies that are performed on awake participants (49). Given these unique strengths, fNIRS has been used to study FC in longitudinal studies in awake infants in high-resource settings (31), and the portability and the low cost of this equipment has fostered implementation outside conventional labs and in LMICs (50). Because fNIRS overcomes logistical (MRI) and resolution (EEG) challenges of other common modalities, we chose it for this study to assess early life brain health of children in LMIC exposed to early severe undernutrition.

The goal of the study was to investigate early physical growth in infancy, developmental trajectories of brain FC across the first two years of life, and cognitive outcome at school age in a longitudinal cohort of infants and children from rural Gambia, an environment with high rates of maternal and child undernutrition. Specifically, we aimed to: (i) investigate whether differences in physical growth through the first two years of life are related to FC at 24 months, and (ii) investigate if early FC has an impact on cognitive outcome at pre-school age in these children. Data were collected from N=204 children as part of the Brain Imaging for Global Health project (BRIGHT; globalfnirs.org/the-bright-project), a longitudinal study examining infant development from term birth to 24-months of age. We acquired longitudinal measurements of brain FC in awake Gambian infant participants at five age points (5, 8, 12, 18, and 24 months) using fNIRS as the infants watched calming videos. We also acquired anthropometric measures (i.e., weight and length) as indicators of nutritional status at birth, at 7-to-14 days, one month of age, and at all five fNIRS data acquisition time points. Additionally, because developmental delays interfere with everyday functioning starting within the pre-school age range (16), we assessed cognitive flexibility, a core executive function (51), at ages three and five years. We hypothesized that (i) long-range FC would increase and short-range FC would decrease throughout the first two years of life, (ii) that early positive physical growth would significantly predict long-range FC, and that (iii) long-range FC strength would predict cognitive flexibility performance in pre-schoolers. Our results provide novel insights into the developmental trajectory of the functional organization of the developing brain, its relationship with nutritional adversity exposure and status, and the subsequent effects on preschool-age cognitive outcomes.

Results

Gambian infants exhibit robust FC development throughout first two years

The repeated assessments of brain FC throughout the first two years of life allowed us to analyse developmental trajectories of the maturation of early brain networks and how they relate to measures of infants' growth as well as to cognitive flexibility at preschool age (Figure 1A). To estimate how FC changed between 5-24 months of age in Gambian children, we performed linear mixed models (LMM) using data from the 132 infants with valid data for at least two time points.

To increase statistical power while minimizing errors due to variability in head anatomy and cap placement, we divided the array into six sections and calculated functional connectivity (FC) between all possible 21 interhemispheric homotopic, intrahemispheric, fronto-posterior, and crossed connections (**Figure S11**). Results on the Fisher-z transformed correlation coefficients (z-RHO) of the oxygenated haemoglobin (HbO₂) showed that frontal interhemispheric homotopic FC decreased with age ($F=11.03$, $p<0.001$), and that left ($F=5.8$, $p<0.001$) and right fronto-middle ($F=4.86$, $p<0.001$) and right fronto-posterior ($F=5.52$, $p<0.001$) FC increased with age (**Figure 2**). Results on the z-RHO scores of the deoxygenated haemoglobin (HHb) largely agreed with the results on HbO₂ (**Table 1** and **Figure 2**). Frontal interhemispheric FC significantly decreased with age, showing positive correlations at five months, and negative values at 24 months (**Figure 2B** and **Figure 2C**). The FC between fronto-middle and frontal-posterior that significantly increased with age showed strong anticorrelation values at early ages that gradually decreased but were still negative at 24 months (**Figure 2B** and **Figure 2C**). Notably, results from LMMs performed on data pre-processed without the global signal regression (GSR) (see results in SI, **Table S11** and **Figure S12**) confirmed a decreasing trajectory of the frontal interhemispheric homotopic FC and an increasing trajectory of bilateral short-range FC with age.

Early growth trajectories predict functional connectivity at 24 months

To investigate the impact of early nutritional status on FC at 24 months, we used multiple regression with the infant growth trajectory (delta weight for length z-score between all time points, Δ WLZ) and FC at 24 months, adjusting for WLZ at birth or head-circumference z-score (HCZ) at 7/14 days. To maximise power, we considered only those FC that showed a statistically significant change with age. The Δ WLZ between birth/1 month and older ages positively predicted frontal interhemispheric homotopic FC at 24 months (Δ WLZ birth with older ages all $p<0.02$ FDR corrected, Δ WLZ 1 month with older ages all $p<0.03$, see **Figure S13** for an example scatterplot) and negatively predicted left and right fronto-middle FC at 24 months ($p<0.04$). The Δ WLZ between birth and 1 month positively predicted right fronto-posterior FC at 24 months ($p=0.006$, FDR corrected) while Δ WLZ between 1 month and older ages negatively predicted right fronto-posterior FC at 24 months (all $p<0.01$, FDR corrected). Interestingly, Δ WLZ between 5/8/12/18 months and older ages did not show statistically significant impacts on any of the FC assessed (**Table 2** and **Figure 3A**).

Early functional connectivity predicts cognitive flexibility at preschool age

To investigate whether FC early in life predicted cognitive flexibility at preschool age, we used multiple regression of FC across the first two years of life against later cognitive flexibility in preschoolers at three and five years. As per the analysis above, we focused on only those FC that showed a statistically significant change with age. Our results showed that frontal interhemispheric homotopic FC at 5 months ($F(1,38)=4.21$, $p=0.047$, $R^2=0.102$), left frontal-middle ($F(1,33)=7.86$, $p=0.009$, $R^2=0.197$) and right frontal-posterior FC at 12 months ($F(1,38)=4.82$, $p=0.034$, $R^2=0.115$) positively predicted cognitive flexibility in young preschoolers. Right frontal-posterior FC at 8 months negatively predicted cognitive flexibility in young preschoolers ($F(1,32)=5.6$, $p=0.024$, $R^2=0.153$). Frontal interhemispheric homotopic FC at 18 months ($F(1,45)=4.82$, $p=0.030$, $R^2=0.115$), and right frontal-posterior FC at 24 months ($F(1,49)=4.85$, $p=0.032$, $R^2=0.092$) positively predicted cognitive flexibility in older preschoolers. Left front-middle FC at 18 months negatively predicted cognitive flexibility in older preschoolers ($F(1,45)=5.72$, $p=0.021$, $R^2=0.115$). While these associations between early functional connectivity and cognitive flexibility at preschool age show promise, none of these survived FDR correction for multiple comparisons (**Figure 3B**), and so should be considered as preliminary and useful for hypothesis generation for future studies.

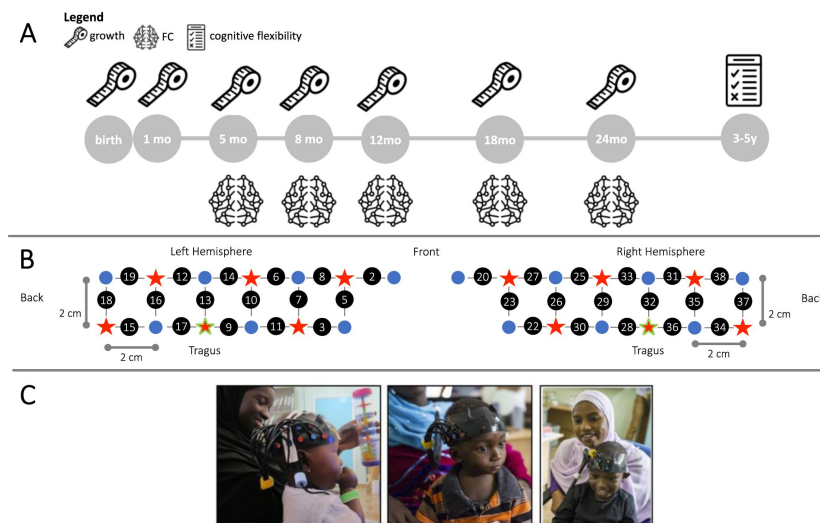


Figure 1.

Experimental design.

(A) Measures taken in the BRIGHT project used in this work. The measuring tape represents anthropometric measures, the brain represents fNIRS FC and the test represents the cognitive flexibility assessment. (B) Schematic representation of the spatial layout of the fNIRS array. Sources are marked with red stars, detectors are marked with blue circles, channels are marked with grey lines and numbered with black circles. The channels/optodes used as a reference for the tragus are highlighted in green. (C) Participants undergoing fNIRS testing at different ages. Image reproduced with permission from (41 [41](#)).

Figure 2.

Linear mixed models results showing FC that displayed a statistically significant change with age.

(A) Significant results of the linear mixed model, blue indicates connections that decreased with age, red indicates connections that increase with age. (B) Mean and standard error of the mean (SE) of the functional connections that changed with age. Error bars are 1 SE. (C) Violin plot showing the mean $\pm$ standard deviation (SD) (red circles and lines) and the individual variability (coloured dots) of the FC that showed a change with time.

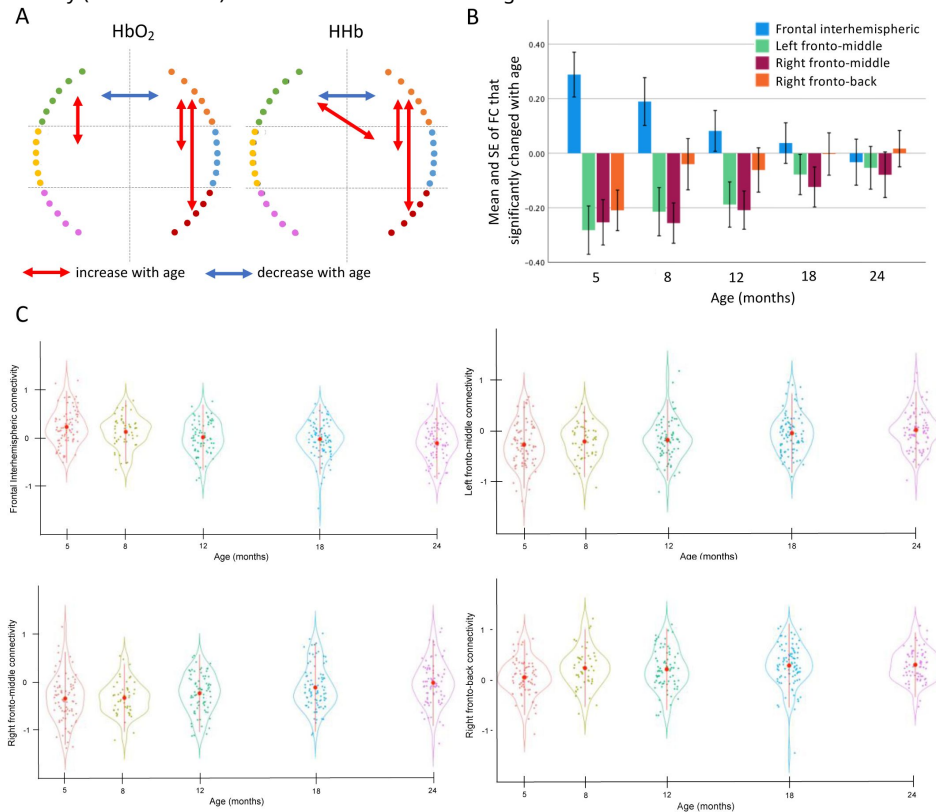


Table 1

FC that significantly changed with age.

Results are displayed in terms of estimated betas, standard errors and p values that survived Bonferroni correction. Regressions that survived Bonferroni correction for multiple comparisons are in bold.

FC	F	p	Baseline (5 months) Betas (SE), p	5-8 change Betas (SE), p	5-12 change Betas (SE), p	5-18 change Betas (SE), p	5-24 change Betas (SE), p
HbO₂							
Frontal interhemispheric	11.03	<0.001	0.28 (0.03), <0.001	-0.09 (0.06), 0.12	-0.20 (0.05), <0.001	-0.25 (0.05), <0.001	-0.33 (0.05), <0.001
Left fronto-middle	5.8	<0.001	-0.28 (0.04), <0.001	0.06 (0.06), 0.293	0.09 (0.05), 0.102	0.19 (0.05), <0.001	0.23 (0.05), <0.001
Right fronto-middle	4.86	<0.001	-0.70 (0.29), 0.018	0.01 (0.05), 0.982	0.05 (0.05), 0.288	0.13 (0.05), 0.007	0.19 (0.05), <0.001
Right frontal-posterior	5.52	<0.001	-0.20 (0.03), <0.001	0.16 (0.06), 0.005	0.14 (0.05), 0.006	0.20 (0.05), <0.001	0.22 (0.05), <0.001
HHb							
Frontal interhemispheric	6.59	<0.001	0.32 (0.03), <0.001	-0.09 (0.06), 0.107	-0.22 (0.05), <0.001	-0.23 (0.05), <0.001	-0.22 (0.05), <0.001
Cross left fronto-right middle	5.17	<0.001	-0.32 (0.04), <0.001	0.06 (0.06), 0.328	0.07 (0.05), 0.188	0.19 (0.05), <0.001	0.22 (0.05), <0.001
Right fronto-middle	5.55	<0.001	-0.78 (0.32), 0.036	0.01 (0.06), 0.86	0.05 (0.05), 0.301	0.16 (0.05), 0.002	0.21 (0.05), <0.001
Right frontal-posterior	5.38	<0.001	-0.24 (0.04), <0.001	0.10 (0.06), 0.073	0.08 (0.05), 0.133	0.23 (0.05), <0.001	0.02 (0.05), <0.001

Frontal interhemispheric FC at 24 mo							
Δ WLZ	birth	1 mo	5 mo	8 mo	12 mo	18 mo	24 mo
birth		NS	$F(1,68)=8.69$, $p=0.004^*$, $R^2=0.102$	$F(1,67)=7.65$, $p=0.007^*$, $R^2=0.104$	$F(1,69)=10.4$, $p=0.002^{**}$, $R^2=0.121$	$F(1,68)=10.98$, $p=0.001^{**}$, $R^2=0.142$	$F(1,59)=5.40$, $p=0.024^*$, $R^2=0.085$
1 mo			$F(1,77)=5.59$, $p=0.021^{**}$, $R^2=0.069$	NS	$F(1,78)=4.74$, $p=0.032^{**}$, $R^2=0.058$	$F(1,75)=4.45$, $p=0.038^{**}$, $R^2=0.057$	NS
5 mo				NS	NS	NS	NS
8 mo					NS	NS	NS
12 mo						NS	NS
18 mo							NS
24 mo							
Left fronto-middle FC at 24 mo							
Δ WLZ	birth	1 mo	5 mo	8 mo	12 mo	18 mo	24 mo
birth		NS	NS	$F(1,67)=4.07$, $p=0.048^{**}$, $R^2=0.058$	NS	NS	$F(1,59)=4.24$, $p=0.044^{**}$, $R^2=0.068$
1 mo			NS	NS	NS	NS	NS
5 mo				NS	NS	NS	NS
8 mo					NS	NS	NS
12 mo						NS	NS
18 mo							NS
24 mo							
Right fronto-middle FC at 24 mo							
Δ WLZ	birth	1 mo	5 mo	8 mo	12 mo	18 mo	24 mo
birth		NS	NS	$F(1,67)=4.74$, $p=0.033$, $R^2=0.067$	NS	$F(1,68)=5.29$, $p=0.024^{**}$, $R^2=0.073$	NS
1 mo			$F(1,77)=4.26$, $p=0.042^{**}$, $R^2=0.053$	$F(1,75)=6.66$, $p=0.012^{**}$, $R^2=0.083$	NS	$F(1,75)=5.33$, $p=0.024^{**}$, $R^2=0.067$	NS
5 mo				NS	NS	NS	NS
8 mo					NS	NS	NS
12 mo						NS	NS
18 mo							NS
24 mo							
Right frontal-posterior FC at 24 mo							
Δ WLZ	birth	1 mo	5 mo	8 mo	12 mo	18 mo	24 mo
birth		$F(1,66)=8.26$, $p=0.006^{**}$, $R^2=0.113$	NS	NS	NS	NS	NS
1 mo			$F(1,69)=4.86$, $p=0.031^*$, $R^2=0.067$	$F(1,68)=6.98$, $p=0.01^{**}$, $R^2=0.094$	$F(1,71)=8.86$, $p=0.004^{**}$, $R^2=0.112$	$F(1,67)=7.72$, $p=0.007^{**}$, $R^2=0.105$	$F(1,57)=7.03$, $p=0.01^{**}$, $R^2=0.112$
5 mo				NS	NS	NS	NS
8 mo					NS	NS	NS
12 mo						NS	NS
18 mo							NS
24 mo							

Table 2.

Results of the regression analyses of the effect of Δ WLZ on FC at 24 months.

Significant positive associations are in green, significant negative associations are in orange, and non-significant (NS) associations are in blue; * indicates regressions that are still significant after correcting for HCAZ at 7/14 days and ** indicates regressions that are still significant after correcting for neonatal HCAZ and WLZ. Regressions that survived FDR correction for multiple comparisons are in bold.

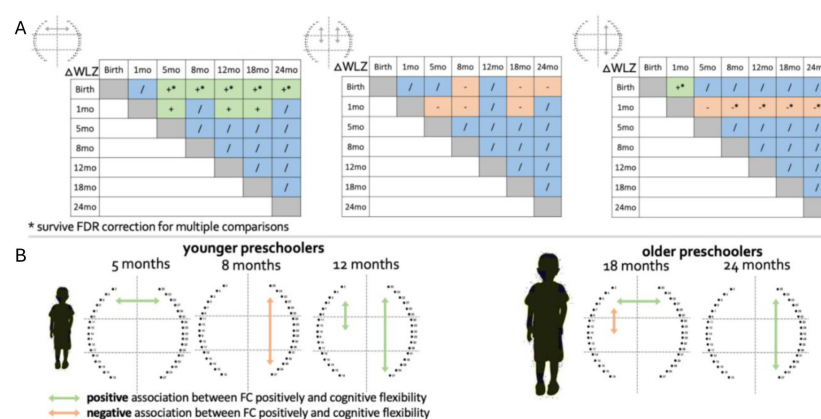


Figure 3.

Associations between FC, early growth and later cognitive flexibility.

(A) Significant positive associations are in green, significant negative associations are in orange, and non-significant associations are in blue. * indicates regressions that survived FDR correction for multiple comparisons. (B) Schematic representations of the early FC connections shown to predict cognitive flexibility in preschoolers. Significant positive associations are in green, significant negative associations are in orange.

Discussion

This study investigated how early adversity via undernutrition drives brain functional connectivity throughout the first two years of life and how these early functional connections are associated with cognitive flexibility at preschool age. To capture the rapid neural development that takes place during early years in community LMIC settings we used fNIRS to assess brain connectivity (22, 24, 26, 31). Additionally, we recorded brain activity while infants were awake, thus making the results more comparable with those from adult studies. Our results show: (i) healthy growth trajectories during early infancy are crucial for healthy FC at 24 months; (ii) our cohort of Gambian infants exhibit atypical developmental trajectories of functional connectivity (FC) in over the first two years of life; and (iii) FC during the first two years of life may predict cognitive flexibility at preschool age.

First, we found that while left and right intra-hemispheric fronto-middle and right frontal-posterior FC increased with age, frontal inter-hemispheric FC decreased with age. Additionally, we observed long-range right frontal-posterior FC increased with age and exhibited positive correlations at 24 months, as consistent with previous infant studies (29, 31). Previous findings from typically developing infants measured in the US showed that long-range FC gradually strengthened with age, indicating maturing of functional networks that span distant brain regions (28, 29). Moreover, the presence of inter-hemispheric connections are known to indicate healthy development of FC (52) consistent with the continuous development of the corpus callosum from infancy until early adulthood (53).

Conversely, decreased inter-hemispheric FC has been observed where development can follow a more divergent and/or heterogeneous path, such as in autism (54). The observed decrease in frontal inter-hemispheric FC with increasing age may be due to the exposure to early life undernutrition adversity. We acknowledge that differences in FC could also be attributed to other environmental and cultural disparities between high-resource and low-resource settings, and future studies are needed to explore this further. Finally, our results indicated that the Gambian infant bilateral fronto-middle FC exhibited anticorrelation at five months, and then gradually became less negative with age. This pattern suggests that this fronto-middle FC may become in sync later after the second year of life. The fNIRS array covers regions over the front and middle cortical regions, putatively belonging to the default mode network and fronto-parietal network, which are normally anticorrelated (55), suggesting that the earlier observed anticorrelation pattern might indeed be adaptive. While further investigations are needed to confirm these findings, these results highlight a general disruption of the integration/segregation process of network development among the infants in our study (56).

Second, a central goal of the current work was to investigate whether infant nutritional status, as measured by change in weight for length z-scores (Δ WLZ), was associated with functional connectivity at 24 months. First, we observed that Δ WLZ between neonatal time points and older ages positively predicted frontal interhemispheric homotopic FC at 24 months, which supports the idea that the decreasing interhemispheric connectivity observed with age might be atypical, as greater increases in Δ WLZ were found to be positively associated with stronger interhemispheric FC. This is consistent with research showing an impact of nutritional status on long-range connections and brain networks development (57–59) and the decreased long-range FC found in preterm babies with faltering growth (60–62). A common feature of the observed associations between Δ WLZ and FC is that they were statistically significant only when Δ WLZ was calculated including measures collected neonatally (i.e., at birth or one month of age). Changes in growth later in development (i.e. Δ WLZ between 5, 8, 12, or 18 months and older ages) did not show statistically significant impacts on any of the FC assessed. This provides support for the hypothesis that undernutrition during the first months of life is more impactful on brain

development than undernutrition occurring later in infancy and early childhood. While the impact of undernutrition on brain development has been documented in LMICs (46), herein, we provided empirical evidence that growth faltering specifically in infants younger than five months of age impacts observable development of functional brain networks in the second year of life. This result also suggests that early undernutrition has a lasting impact on subsequent cognitive skills, even in children who show subsequent catch-up growth.

The other primary aim of this study was to investigate whether early FC development was related to cognitive flexibility in preschoolers at three and five years of age. Long range FC (interhemispheric homotopic and frontal-posterior) positively predicted performance in the cognitive flexibility task both in younger and older preschoolers. This is consistent with the fact that the integration of distant brain regions fosters cognitive development, especially in prefrontal cortex (63, 64), possibly driven by maturation within the fronto-parietal network that has been widely documented as a primary neural underpinning of cognitive flexibility (65). Additionally, left fronto-middle FC at 12 months positively predicted cognitive flexibility in young preschoolers, but the same connection at 18 months negatively predicted cognitive flexibility in older preschoolers. This shift in direction of the relationship between FC and cognitive flexibility might indicate the shift between short-range and long-range FC during development, where short-range FC still promotes cognitive development early in life, while becoming less impactful as the child ages possibly because children develop a wider range of strategies to support cognitive flexibility as they age. A recent review on the neural underpinnings of cognitive flexibility highlighted that right-lateralized activations underlying this skill increased with age, while activations over the left hemisphere decreased with age (51). Our findings are consistent with this view, suggesting an increase in functional specialisation and segregation with development. While our results are consistent with previous studies, we acknowledge that the significant association between early FC and later cognitive flexibility does not withstand multiple comparisons. Therefore, we encourage future studies that may replicate these findings with a larger sample.

Some limitations of the current study merit discussion. Although the BRIGHT sample is one of the largest longitudinal neuroimaging studies in LMIC to assess longitudinally this many age points during the first two years of life, the sample size may still be underpowered to assess the modest effects of the regression models. Second, additional factors may affect FC that were not considered in our models. Applying more advanced statistical modelling methods and structural equation modelling analyses may provide greater insight with further investigations in contexts of adversity and, in turn, establish which outcomes are predicted by FC. That said, results from the regression models were statistically robust, as most survived FDR correction for multiple comparisons, and were significant after including HCZ at 7-14 days and WLZ at birth in the models. It is also worth noting that in line with previous FC fNIRS studies (31, 66), linear mixed model results on the HbO₂ and the HHb signals showed consistency in the FC changes with age measured in the two chromophores, and similar trajectories when removing the global signal regression step from the pre-processing, suggesting that the results were reliable and statistically robust. Future collaborations between projects studying the impact of adversity on brain development in both high resource and LMIC settings are crucial to advance the field and therefore inform efficient strategies – including the timing – of intervention. Even given the large sample in our study, we were underpowered to test for group comparisons between sets of infants with distinct undernutrition growth profiles, e.g., infants with early poor growth that later resolved and infants with standard growth early that had a poor growth later. We were also underpowered to test the associations between early growth and FC on clinically undernourished infants (defined as having Δ WLZ two standard deviations below the mean). However, our findings provide a solid foundation for future research on the impacts of undernutrition during the first months of life on later brain health and cognitive abilities throughout childhood.

In conclusion, herein we showed that, as expected, positive growth trajectories during the first few months of life were positively associated with stronger interhemispheric frontal FC and negatively associated with bilateral short-range FC. Interestingly, while Gambian infants had expected increasing left and right intra-hemispheric fronto-middle and right frontal-posterior FC with age, frontal inter-hemispheric FC decreased with age, which is inconsistent with extant data from other populations studied in Europe and US. Of particular note, growth during the first few months after birth showed a significant impact on brain health, as measured with FC, while later growth changes were not related to variability in FC. Moreover, long-range FC predicted the performance on a cognitive flexibility task at preschool age. Taken together, our results show the impact that early growth has on functional brain development, which in turn affects cognitive flexibility in preschoolers. Finally, this study emphasizes the importance of the timing of interventions, for which benefits are visible at preschool age and beyond.

Materials and Methods

Experimental Design

Our study investigated the developmental trajectory of brain health as assessed with functional connectivity (FC) in Gambian infants and its relationship with early growth, as a proxy for nutritional status, and later cognitive outcomes. To do, so we estimated FC from data acquired with fNIRS longitudinally at five time points between age five and 24 months in the context of the BRIGHT Project. The BRIGHT project sought to provide brain function-for-age curves from the UK and Gambian infants and to gain insight into the effects that issues related to living in low-resource settings may have on brain development. We then tested for relationships between brain FC at age 24 months with measures of early growth, as indexed by changes in weight-for-length z-scores (reflecting body weight in proportion to attained growth in length) at one month of age, and at each of the four subsequent visits (details provided below). We also tested for associations between brain FC at each time point with later cognitive outcome, as indexed by cognitive flexibility assessed in three- and five-year-olds preschool-age children ([Figure 1A](#)). Ethical approval was granted by the joint Gambia Government - MRC Unit the Gambia Ethics Committee, and written informed consent was obtained from all parents prior to participation.

Participants

We used data from N=204 infants recruited in rural community settings of The Gambia as part of the BRIGHT project. The study was a prospective, longitudinal study with fNIRS assessments planned for awake infants at ages 5, 8, 12, 18 and 24 months. All infants were born full term (37–42 weeks gestation). fNIRS data were additionally acquired from 1-month-old infants while asleep. However, as sleep stages may impact connectivity measures (98), these data are not part of the current analysis. Reasons for exclusion from the fNIRS analysis were: i) fussiness; ii) experimental errors (missing pictures of the headgear placement or any other technical issue); iii) poor headgear placement (see supplementary information for *Assessment of headgear placement*); iv) low-quality data (more than 40% of the channels excluded as previously done ([67](#)–[70](#))). [Table S12](#) summarises the details of the included and excluded participants for each visit, showing an attrition rate within the standard range for infant fNIRS studies ([71](#)) (see [Table 3](#) for the demographic information of the participants included in the analyses at each age).

Anthropometric measures

Measurements of length, weight, and head circumference (HC) were collected on all infants at birth, at 7/14 days, 1 month, and in all visits from 5 to 24 months, by trained field workers using calibrated tools. Length was measured using a Harpenden Infantometer length board (Holtain Ltd) to a precision of 0.1 cm. Weight was obtained using an electronic baby scale (Model 336, SECA) to a

Age	Age in days (mean±SD)	Sex (M, F)	WLZ (mean±SD)	HCZ (mean±SD)
5 months	158.92±9.91	45,42	-0.24±0.95	-0.72± 0.92
8 months	249.34±17.97	28,25	-0.24±1.01	-0.81±0.90
12 months	372.92±14.75	42,40	-0.58±1.05	-0.95±0.95
18 months	560.37±24.38	48,49	-0.85±0.91	-0.82±0.93
24 months	746.61±23.25	49,47	-0.44±0.99	-0.80±0.97

Table 3.

Demographic information per each age.

precision of 0.01 kg. Finally, HC was measured around the maximum circumference of the head (forehead to occiput) using stretch-proof measuring tape (Model 201, SECA) to the nearest 0.1 cm. Each measure was taken in triplicate and the mean of the three measures was used in analyses.

fNIRS data acquisition

fNIRS data were collected using the NTS topography system (Gowerlabs Ltd., London, UK). This system uses two continuous wavelengths of near-infrared light (780 and 850 nm) to detect changes in HbO₂ and HHb concentrations, using a sampling rate of 10 Hz (72 [↗](#)).

Infants were assessed using a custom-built fNIRS headgear consisting of two arrays, including 14 sources and 12 detectors to create a total of 34 channels, covering bilateral frontal, inferior-frontal and temporal regions, with a source-detector separation of 2 cm (Figure 1B [↗](#) and Figure 1C [↗](#)). Infants underwent several fNIRS tasks as part of the BRIGHT project, focusing on social processing, deferred imitation, and habituation response. Therefore, brain regions for data acquisition were chosen to maximize the likelihood of recording meaningful data for all tasks (73 [↗](#)).

For each infant, head measurements were taken to enable the alignment of the headgear with the 10–20 placement coordinates (74 [↗](#)). Over the front of the head, the headgear was placed so that a vertical red line marking the centre of the band aligned with the nasion with the silicon band laying just above, and in line with, the eyebrows. The headgear was placed so that the reference optode (the third lower optode from the back) was placed over the tragus.

To model for the headgear placement offline, photographs of the participants (one from the front and one from either side) were taken after the headgear was placed and again at the end of the fNIRS session. To assess the headgear placement, photographs of the participants had to be sufficiently good in quality (i.e. not blurred) and with the camera pointing at the ear, at a position directly perpendicular to the infant's anatomical reference on the side of the head being photographed.

Infants were excluded from further analysis if the band was excessively high over the front above the eyebrows, by checking the displacement of the lower edge of the band with respect of the eyebrows with a “traffic-light” code (red, orange, green). Three experimenters independently classified the infants' frontal headgear placement, and discrepancies were discussed until agreement was reached. Channels of those infants with valid placement over the front but horizontal lateral displacement between the ideal and the actual position of the reference (optode marked in green in Figure 1B [↗](#)) equal or greater than 1.6 cm were renumbered, so that each channel was shifted either backward or forward one full channel location in space

At each visit, the fNIRS data acquisition took place while infants sat on their parent's lap. The parent was instructed to refrain from interacting with the infant during the stimuli presentation. Videos were of male and female Gambian adults singing nursery rhymes in Mandinka (for 70 seconds) and videos of toys in action (for 60 seconds), video blocks alternated, with each type shown to the infant twice, with the aim of keeping the infant awake, calm, attentive and as still as possible. The functional connectivity (FC) paradigm formed part of a greater battery of fNIRS paradigms that ran in a continuous and consistent order across all infants. The fNIRS FC acquisition lasted for a maximum of 520 seconds (two rounds of 260 seconds each) or were stopped as soon as the participant became fussy. Stimuli were presented using a MATLAB® custom-written stimulus presentation framework, Task Engine (sites.google.com/site/taskenginedoc) and Psychtoolbox on an Apple Macintosh computer.

fNIRS preprocessing and data-analysis

Data analyses were carried out using in-house programmes developed in MATLAB (MathWorks, Natick, MA) (**Figure S14** [summarises fNIRS data preprocessing steps](#)). Stringent multi-level data-quality assessments were used to ensure all data included in the group were of adequate quality. First, low-quality channels based on physiological indicators of quality were pruned using QT-NIRS (<https://github.com/lpollonini/qt-nirs> [\(76\)](#)). The software takes advantage of the fact that the cardiac pulsation is also recorded in addition to haemodynamic activity associated with brain-driven variance including task-based responses and those contributing to functional connectivity [\(75\)](#), and quantifies its strength in the spectral and temporal domains with two measures: the Scalp Coupling Index (SCI) and the Peak Spectral Power (PSP) (for more details, see [\(76\)](#)). For each infant, data quality was assessed channel-by-channel, and SCI and PSP were calculated with sliding 3-second windows (threshold SCI=0.70, threshold PSP=0.1, empirically defined and based on [\(77\)](#)) [\(76\)](#). Channels that had both SCI and PSP below threshold for more than 70% of the windows [\(76\)](#), were excluded from further pre-processing steps.

Raw intensity data were converted to optical density via a logarithm after dividing by the temporal mean of each channel (*hmrIntensity2OD.m* function from Homer2 tool [\(78\)](#)) and bandpass filtered (0.009 – 3 Hz) (*hmrBandpassFilt.m* function from Homer2 [\(78\)](#)). Using first a wide filter leaves in the signal contamination from the high bandwidth information (i.e. respiration, pulse, etc.) that we aim to regress out next. To help manage the systemic arterial signal that is well known to contaminate fNIRS signals [\(79\)](#), we regressed out the mean of the signals across the array, a method also known as global signal regression (GSR) [\(80, 81\)](#) (Results of analysis performed also without GSR are reported in the Supplementary Materials). Hereafter, optical density data underwent a second bandpass filtered (0.009 – 0.08 Hz), as previously described [\(47\)](#). Motion artefacts were then rejected using global variance of temporal derivatives (GVTD, STD=5) in the NeuroDOT toolbox (<https://www.nitrc.org/projects/neurodot/>) [\(82\)](#) (**Figure S15** [show how we tested the effect of different STD thresholds of GVTD on data inclusion](#)). To further manage potential motion contamination of the data, we additionally removed five seconds before and after each motion artefact, and only chunks of data that were at least 20 seconds long were retained.

Optical density data were converted to relative concentrations of haemoglobin using the modified Beer–Lambert law (*hmrOD2Conc.m* function from Homer2 [\(78\)](#)). Differential path length factors were calculated [\(83\)](#) and adapted to wavelengths (780 and 850 nm) and ages (5 months = 5.24 and 4.25; 8 months = 5.26 and 4.26; 12 months = 5.27 and 4.28 ;18 months = 5.30 and 4.30; 24 months = 5.32 and 4.32).

To increase the statistical power of our analysis and to reduce the number of multiple comparisons for all statistical analysis to investigate developmental changes in connectivity, we averaged the concentration changes of the channels that survived pre-processing into sections. The sections were established via the 17 channels of each hemisphere which were grouped into front, middle and back (for a total of six regions) based on a previous co-registration of the BRIGHT fNIRS arrays onto age-appropriate templates [\(68\)](#) (**Figure S11** [\(68\)](#)). Each section represented a reasonable estimate of an anatomically consistent array coverage across the group, and was used to increase statistical power while minimizing errors due to variability in head anatomy and cap placement. For each participant, the Pearson-r correlation matrix between all the sections was calculated for both HbO₂ and HHb, resulting in a 6×6 matrix of section-pair correlations. We then applied a Fisher z-transformation on the correlation matrix for further statistical analyses. Infants with at least 250 seconds of valid data after pre-processing were considered for further analyses. To choose the optimal minimum amount of valid data, per each infant at each time point for both the HbO₂ and HHb signal, we correlated the connectivity matrices estimated in the first portion of data with the one in the last portion of data by increasing the seconds of data considered (i.e. the first 60 seconds with the last 60 seconds, the first 100

seconds with the last 100 seconds, the first 120 seconds with the last 120 seconds, etc.). We found that correlating the first and the last 250 seconds of valid data after pre-processing provided the highest percentage of infants with strong correlation between the first and the last portion of data, suggesting that FC patterns reached a usable stability after 250 seconds (Figure S16).

Cognitive flexibility at preschool age

Gambian infants from BRIGHT were cross-sectionally assessed at the age of 3 or 5 years for cognitive flexibility using the “card sorting” task from the tablet-based Early Years Toolbox (84) (<http://www.eytoolbox.com.au>). In this task, children were asked to sort cards (i.e. red rabbits or blue boats) either by colour (red or blue) or shape (rabbits or boats), and to flexibly switch back and forth between these rules (85).

Due to disruptions to field work as a result of a political crisis in December 2016 - January 2017 that forced us to pause recruitment, data were collected in two ranges of preschool ages, younger preschoolers (age mean $\pm$ SD=47.96 $\pm$ 2.77 months, N=77) and older preschoolers (age mean $\pm$ SD=57.58 $\pm$ 2.11 months, N=84) (73).

Statistical Analyses

To investigate developmental trajectories of functional connectivity between 5 and 24 months, we ran linear mixed models (86) which is the standard statistical test to analyse repeated measures and longitudinal data with dropouts (87). Statistical analysis was performed with SPSS Version 28.0 statistic software package. Compared to repeated measures ANOVA, linear mixed models account for within-participant dependence and allow for missing data by using only information from the individual at the other visits (88). All possible interhemispheric homotopic, intrahemispheric, fronto-posterior, and crossed connections between the six regions were inserted as dependent variables in a linear mixed model, for a total of 21 linear mixed models. Functional connectivity was modelled as the dependent variable as a function of age with a random participant effect and random errors.

Intercept and age were fixed effects, while within-participant dependence was modelled as a random effect (89). This same procedure was used in other longitudinal studies that explored developmental changes over time (31, 90, 91). The linear mixed model included the 132 infants who had valid data for at least two visits (55 infants contributed with data from two visits, 51 infants contributed with data from three visits, 22 infants contributed with data from four visits, four infants contributed with data from all of the five visits). The type of covariance between the observations was specified as Autoregression (AR) as two measures close in time of the same participant are likely to be correlated (92). To ensure statistical reliability, significant results from the linear mixed models were corrected for multiple comparisons using Bonferroni correction ($p=0.00238$).

Anthropometric measures were converted to age and sex adjusted z-scores that are based on World Health Organization Child Growth Standards (93). Weight-for-Length (WLZ) and Head Circumference (HCZ) z-scores were computed. Delta WLZ (Δ WLZ) were calculated for various periods (0-1 months, 0-5 months, etc.) by subtracting the z-score at the later age from that at the previous age. The use of delta z-scores as outcome measures allowed us to assess the impact of positive or negative deviation from the expected growth trajectories. To investigate the impact of undernutrition on FC development, we used Δ WLZ as independent variables in regression analyses on HbO₂ (as the chromophore with the highest signal-to-noise ratio) FC at 24 months, our final time point of data collection. To maximise power, we considered only those FC that showed a statistically significant change with age. These analyses were adjusted by WLZ at birth and HCZ at 7/14 days, to more confidently assume that the associations between growth and FC were specific to the impact of change in WLZ postnatally and not confounded by the size or maturity of the infant at birth. We used HCZ measured at 7/14 days and not HCZ at birth as HC measures at birth

are unreliable following vaginal delivery. Therefore, HCZ at 7/14 days is a more accurate head measure than HCAZ at birth. To ensure statistical reliability, results from the regression analyses on each FC were corrected for multiple comparisons using false discovery rate (FDR)(94 [↗](#)) per each connection investigated, i.e. 21 possible Δ WLZ values per each connection.

To investigate whether FC early in life predicted cognitive flexibility at preschool age, we regressed FC that showed a significant change across the first two years of life against later cognitive flexibility.

Data and materials availability

Data supporting this paper will be made available subject to established data sharing agreements.

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Author contribution

Conceptualization: CB, SEM, SLF, CEE, ATE

Methodology: CB, AB, SMC, BM, SEM, SLF, CEE, ATE

Investigation: SMC, GG, EM, ET, TF, LA

Visualization: CB, ATE

Supervision: SEM, SLF, CEE, ATE

Writing-original draft: CB, ATE

Supplementary Results

Linear mixed model results showing the developmental trajectories of functional connectivity in Gambian infants over the first 2 years of life age (fNIRS pre-processing without global signal regression)

To rule out that the global signal regression (GSR) performed as part of our fNIRS preprocessing might had an impact on our results, we re-processed the data without this step. All the other steps were kept the same. Hereafter, we reperformed the linear mixed models (LMM) on all the possible 21 interhemispheric homotopic, intrahemispheric, fronto-posterior, and crossed connections.

Results on the Fisher-z transformed correlation coefficients (z-RHO scores) on the oxygenated haemoglobin (HbO₂) showed that left (F=10.4, $p<0.001$) and right fronto-middle (F=4.82, $p<0.001$) FC increased with age. Results on the Fisher-z transformed correlation coefficients (z-RHO scores) on the deoxygenated haemoglobin (HHb) showed that showed that frontal interhemispheric FC decreased with age (F=3.5, $p<0.002$) ([Table SI1](#) and [Figure SI2](#)).

Supplementary Figures

Figure SI1.

Schematic representation of the fNIRS array and the functional connections tested.

(A) Each dot represents a channel, colours on the left plot corresponds to colours on the right on the baby's head. 6 sections. (B) The 21 connections tested in the linear mixed models. Interhemispheric homotopic connections are in orange (connecting the same regions between hemispheres, i.e., front left with front right), intrahemispheric connections are in green (correlations of channels belonging to the same region), fronto-posterior are in blue (connecting front and middle, middle and back, and front and back regions of the same hemisphere), and crossing interhemispheric connections (interhemispheric non-homotopic, connecting the front and middle, middle and back, and front and back regions of the two hemispheres) are in yellow.

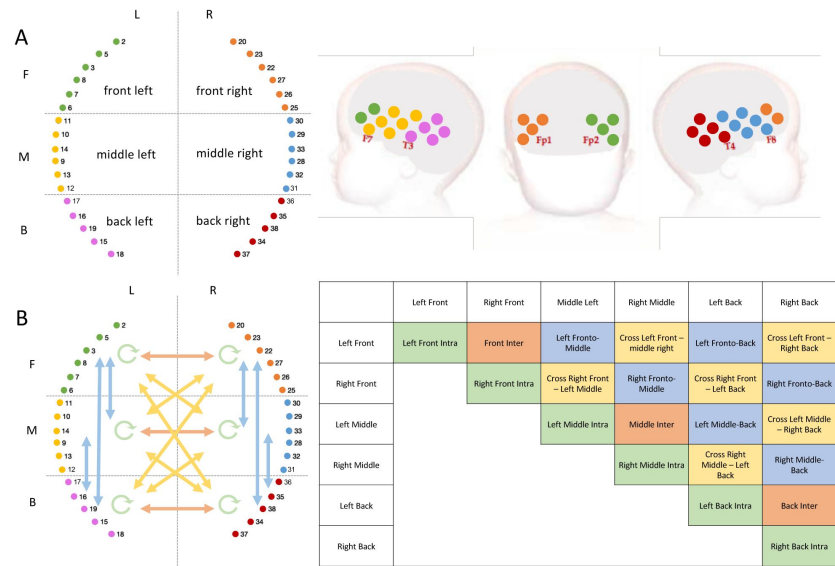


Figure SI2.

Linear mixed models result showing FC that displayed a statistically significant change with age.

(A) Results of the linear mixed model, blue indicates connections that decreased with age, red indicates connections that increase with age. (B) Mean and SE of the functional connections that changed with age. Error bars are 1 SE.

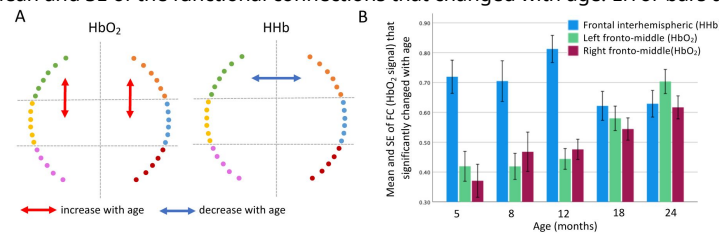


Figure SI3.

An example scatterplot of the association between Δ WLZ and FC at 24 months.

Scatterplot of the association between Δ WLZ between birth and 5 months and frontal interhemispheric connectivity at 24 months. The black line represents the line of best fit.

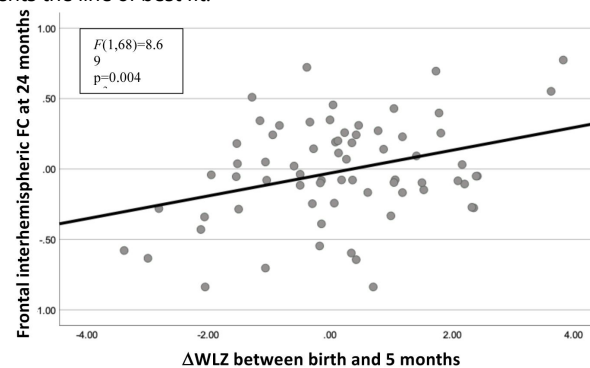


Figure SI4.

fNIRS preprocessing steps.

The column “infants included in the analyses” in [Table 3](#) refers to those participants whose data survived these preprocessing steps.

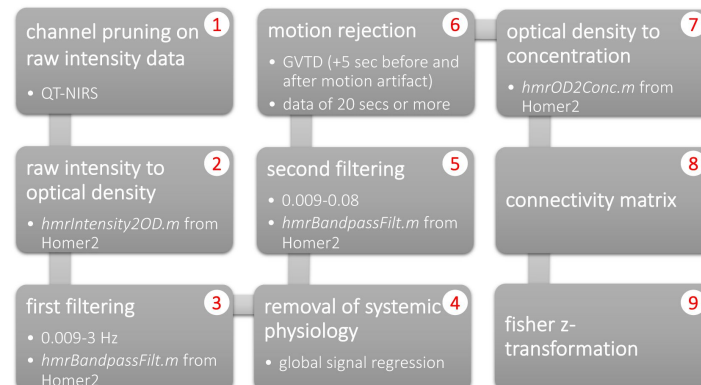


Figure SI5.

The effect of different thresholds for GVTd for motion detection and minimum valid data after pre-processing on data inclusion.

(A) Percentage of data included (left) and seconds of data included (right) per participant. Each line represents an infant, the black line represents the mean value. These graphs are reported from the 12 months sample as example. (B) Number of infants included in the LMM by varying STD threshold for GVTd and minimum length of valid data at different ages.

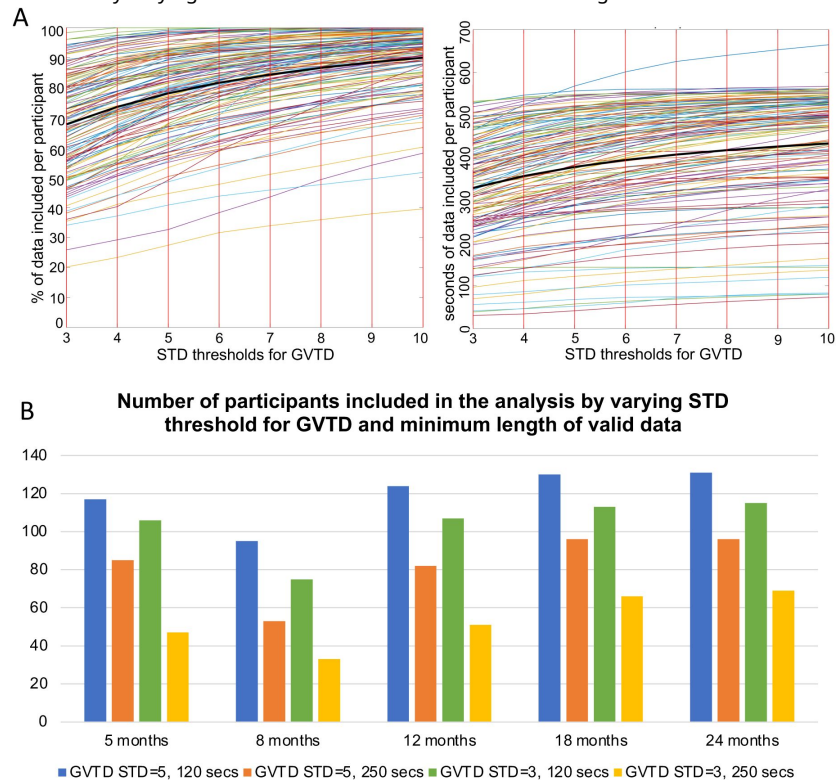
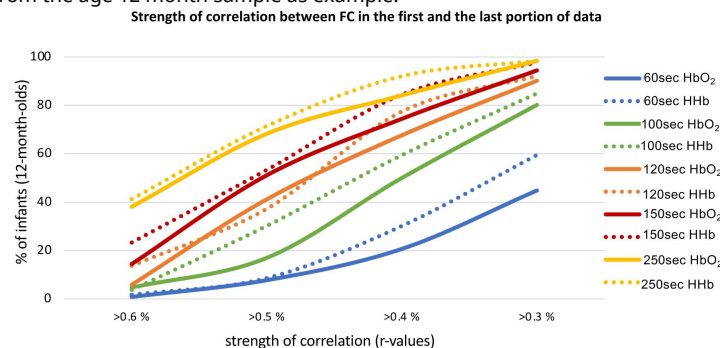


Figure SI6.

Strength of correlation between FC in the first and last portion of data.

This graph is reported from the age 12 month sample as example.



Supplementary Tables

Table SI1.

FC that significantly changed with age (fNIRS pre-processing without global signal regression).

Results are displayed in terms of estimated betas, standard errors and p values.

FC	F	p	Baseline (5 months) Betas (SE), p	5-8 change Betas (SE), p	5-12 change Betas (SE), p	5-18 change Betas (SE), p	5-24 change Betas (SE), p
HbO₂							
Left fronto-middle	10.4	<0.001	-0.55 (0.29), <0.063	0.05 (0.05), 0.239	0.07 (0.04), 0.125	0.16 (0.04), <0.001	0.26 (0.04), <0.001
Right fronto-middle	4.82	<0.001	-0.24 (0.28), 0.398	0.03 (0.05), 0.535	0.05 (0.04), 0.246	0.08 (0.04), 0.051	0.18 (0.04), <0.001
HHb							
Frontal interhemispheric	3.5	0.002	0.18 (0.33), 0.591	-0.07 (0.06), 0.260	-0.07 (0.05), 0.214	-0.17 (0.05), 0.002	-0.17 (0.05), 0.002

Table SI2.

Characteristics of included and excluded participants and seconds of data included in the analyses at each age.

WD=withdrawn, D=deceased, MV=missed visit, DD=developmental delay, NIRS not undertaken = the participant was assessed but did not want or could not perform the NIRS assessments, FC not undertaken = the participant was assessed with other NIRS task, but not FC, Fussied out = the participant wore the headband and the FC acquisition had started but the participant showed signs of fussiness soon after the start of the acquisition, MP=missing pictures of the headband placement, EM=missing event markers, TI=technical issues during the NIRS testing session. The proportion of children included in the analysis was computed based on the infants with FC data.

Age	N	Not tested				NIRS not undertaken	FC not undertaken	Infants with FC data	Fussied out	Experimental errors			Headband Placement	Too many channels excluded	Not enough data after pre-processing	Infants included in the analyses	Seconds of data included in the analyses (mean±SD)	Inclusion rate (from the 204 infants recruited)
		WD	D	MV	DD					MP	EM	TI						
5 months	N	2	1	2	3	10	7	179	16	11	3	3	7	5	47	87	382.68±92.78	42%
	%	0.98	0.49	0.98	1.47	4.90	3.43	87.74	8.93	5.39	1.67	1.67	5.58	2.79	26.25	48.6		
8 months	N	7	1	5	3	18	14	156	7	6	0	1	23	6	60	53	372.93±80.66	25%
	%	3.43	0.49	2.45	1.47	8.82	6.86	76.47	4.48	3.84	0	0.64	14.7	3.84	38.46	33.97		
12 months	N	9	1	4	3	17	13	157	4	1	0	2	10	2	56	82	372.93±80.66	40%
	%	4.41	0.49	1.96	1.47	8.33	6.37	76.96	2.54	0.63	0	1.27	6.36	1.27	35.66	52.22		
18 months	N	8	1	15	3	12	5	160	4	0	0	1	16	4	38	97	388.85±81.63	47%
	%	3.92	0.49	7.35	1.47	5.88	2.45	78.43	2.5	0	0	0.62	10	2.5	23.75	60.62		
24 months	N	13	1	29	1	3	4	153	0	2	0	0	6	4	45	96	399.56±79.86	47%
	%	6.37	0.49	14.2	0.49	1.47	1.96	75	0	1.30	0	0	3.92	2.61	29.41	62.74		

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Reviewer #1 (Public review):

Summary:

This study utilises fNIRS to investigate the effects of undernutrition on functional connectivity patterns in infants from a rural population in Gambia. fNIRS resting-state data recording spanned ages 5 to 24 months, while growth measures were collected from birth to 24 months. Additionally, executive functioning tasks were administered at 3 or 5 years of age. The results show an increase in left and right frontal-middle and right frontal-posterior connections with age and, contrary to previous findings in high-income countries, a decrease in frontal interhemispheric connectivity. Restricted growth during the first months of life was associated with stronger frontal interhemispheric connectivity and weaker right frontal-posterior connectivity at 24 months of age. Additionally, the study describes some connectivity patterns, including stronger frontal interhemispheric connectivity, which is associated with better cognitive flexibility at preschool age.

Strengths:

- The study analyses longitudinal data from a large cohort ($n = 204$) of infants living in a rural area of Gambia. This already represents a large sample for most infant studies, and it is impressive, considering it was collected outside the lab in a population that is underrepresented in the literature. The research question regarding the effect of early nutritional deficiency on brain development is highly relevant and may highlight the importance of early interventions. The study may also encourage further research on different underrepresented infant populations (i.e., infants not residing in Western high-income countries) or in settings where fMRI is not feasible.

- The preprocessing and analysis steps are carefully described, which is very welcome in the fNIRS field, where well-defined standards for preprocessing and analysis are still lacking.

Weaknesses:

- While the study provides a solid description of the functional connectivity changes in the first two years of life at the group level and investigates how restricted growth influences connectivity patterns at 24 months, it does not explore the links between adverse situations and developmental trajectories for functional connectivity. Considering the longitudinal nature of the dataset, it would have been interesting to apply more sophisticated analytical tools to link undernutrition to specific developmental trajectories in functional connectivity. The authors mention that they lack the statistical power to separate infants into groups according to their growing profiles. However, I wonder if this aspect could not have been better explored using other modelling strategies and dimensional reduction techniques. I can think about methods such as partial least squares correlation, with age included as a numerical variable and measures of undernutrition.

- Connectivity was assessed in 6 big ROIs. While the authors justify this choice to reduce variability due to head size and optode placement, this also implies a significant reduction in spatial resolution. Individual digitalisation and co-registration of the optodes to the head model, followed by image reconstruction, could have provided better spatial resolution. This is not a weakness specific to this study but rather a limitation common to most fNIRS studies, which typically analyse data at the channel level since digitalisation and co-registration can be challenging, especially in complex setups like this. However, the BRIGHT project has demonstrated that it is possible and that differences in placement affect activation patterns, which become more localised when data is co-registered at the subject level (Collins-Jones et al., 2021). Could the co-registration of individual data have increased sensitivity, particularly given that longitudinal effects are being investigated?

- I believe that a further discussion in the manuscript on the application of global signal regression and its effects could have been beneficial for future research and for readers to better understand the negative correlations described in the results. Since systemic physiological changes affect HbO/HbR concentrations, resulting in an overestimation of functional connectivity, regressing the global signal before connectivity computation is a common strategy in fNIRS and fMRI studies. However, the recommendation for this step remains controversial, likely depending on the case (Murphy & Fox, 2017). I understand that different reasons justify its application in the current study. In addition to systemic physiological changes originating from brain tissue, fNIRS recordings are contaminated by changes occurring in superficial layers (i.e., the scalp and skull). While having short-distance channels could have helped to quantify extracerebral changes, challenges exist in using them in infant populations, especially in a longitudinal study such as the one presented here. The optimal source-detector distance that minimises sensitivity to changes originating from the brain would increase with head size, and very young participants would require significantly shorter source-detector distances (Brigadoi & Cooper, 2015). Thus, having them would have

been challenging. Under these circumstances (i.e., lack of short channels and external physiological measures), and considering that the amount the signal is affected by physiological noise (either coming from the brain or superficial tissue) might change through development, the choice of applying global signal regression is justified. Nevertheless, since the method introduces negative correlations in the data by forcing connectivity to average to zero, I believe a further discussion of these points would have enriched the interpretation of the results.

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Reviewer #2 (Public review):

Strengths:

The article addresses a topic of significant importance, focusing on early life growth faltering in low-income countries—a key marker of undernutrition—and its impact on brain functional connectivity (FC) and cognitive development. The study's strengths include the laborious data collection process, as well as the rigorous data preprocessing methods employed to ensure high data quality. The use of cutting-edge preprocessing techniques further enhances the reliability and validity of the findings, making this a valuable contribution to the field of developmental neuroscience and global health.

Weaknesses:

The study fails to fully leverage its longitudinal design to explore neurodevelopmental changes or trajectories, as highlighted by all three reviewers. The revised manuscript still primarily focuses on FC values at a single age stage (i.e., 24 months) rather than utilizing the longitudinal data to investigate how FC evolves over time or predicts cognitive development. Although the authors acknowledge that analyzing changes in FC (ΔFC) would reduce degrees of freedom (to ~ 30) and risk interpretability, they do not report or discuss these results, even as exploratory findings.

Furthermore, the study lacks specificity in identifying which specific brain networks are affected by growth faltering, as the current exploratory analyses mainly provide an overall conclusion that infant brain network development is impacted without pinpointing the precise neural mechanisms or networks involved.

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Reviewer #3 (Public review):

Summary

This study aimed to investigate whether the development of functional connectivity (FC) is modulated by early physical growth, and whether these might impact cognitive development in childhood. This question was investigated by studying a large group of infants (N=204) assessed in Gambia with fNIRS at 5 visits between 5 and 24 months of age. Given the complexity of data acquisition at these ages and following data processing, data could be analyzed for 53 to 97 infants per age group. FC was analyzed considering 6 ensembles of brain regions and thus 21 types of connections. Results suggested that: i) compared to previously studied groups, this group of Gambian infants have different FC trajectory, in particular with a change in frontal inter-hemispheric FC with age from positive to null values; ii) early physical growth, measured through weight-for-length z-scores from birth on, is associated with FC at 24 months. Some relationships were further observed between FC

during the first two years and cognitive flexibility, in different ways between 4- and 5-year-old preschoolers, but results did not survive corrections for multiple comparisons.

Strengths

The question investigated in this article is important for understanding the role of early growth and undernutrition on brain and behavioral development in infants and children. The longitudinal approach considered is highly relevant to investigate neurodevelopmental trajectories. Furthermore, this study targets a little studied population from a low-/middle-income country, which was made possible by the use of fNIRS outside the lab environment. The collected dataset is thus impressive and it opens up a wide range of analytical possibilities.

Weaknesses

- Data analyses were constrained by the limited number of children with longitudinal data on NIRS functional connectivity. Nevertheless, considering more advanced statistical modeling approaches would be relevant to further explore neurodevelopmental trajectories as well as relationships with early growth and later cognitive development.
- The abstract and end of the discussion should make it clearer that the associations between FC and cognitive flexibility are results that need to be confirmed, insofar as they did not survive correction for multiple comparisons.

<https://doi.org/10.7554/eLife.94194.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Cognitive and brain development during the first two years of life is vast and determinant for later development. However, longitudinal infant studies are complicated and restricted to occidental high-income countries. This study uses fNIRS to investigate the developmental trajectories of functional connectivity networks in infants from a rural community in Gambia. In addition to resting-state data collected from 5 to 24 months, the authors collected growing measures from birth until 24 months and administered an executive functioning task at 3 or 5 years old.

The results show left and right frontal-middle and right frontal-posterior negative connections at 5 months that increase with age (i.e., become less negative). Interestingly, contrary to previous findings in high-income countries, there was a decrease in frontal interhemispheric connectivity. Restricted growth during the first months of life was associated with stronger frontal interhemispheric connectivity and weaker right frontal-posterior connectivity at 24 months. Additionally, the study describes that some connectivity patterns related to better cognitive flexibility at pre-school age.

Strengths:

- *The authors analyze data from 204 infants from a rural area of Gambia, already a big sample for most infant studies. The study might encourage more research on different underrepresented infant populations (i.e., infants not living in occidental high-income countries).*

- *The study shows that fNIRS is a feasible instrument to investigate cognitive development when access to fMRI is not possible or outside a lab setting.*
- *The fNIRS data preprocessing and analysis are well-planned, implemented, and carefully described. For example, the authors report how the choices in the parameters for the motion artifacts detection algorithm affect data rejection and show how connectivity stability varies with the length of the data segment to justify the threshold of at least 250 seconds free of artifacts for inclusion.*
- *The authors use proper statistical methods for analysis, considering the complexity of the dataset.*

We thank the reviewer for highlighting the strengths of this work.

Weaknesses:

- *No co-registration of the optodes is implemented. The authors checked for correct placement by looking at pictures taken during the testing session. However, head shape and size differences might affect the results, especially considering that the study involves infants from 5 months to 24 months and that the same fNIRS array was used at all ages.*

The fNIRS array used in this work was co-registered onto age-appropriate MNI templates at every time point in a previous published work L. H. Collins-Jones, et al., Longitudinal infant fNIRS channel-space analyses are robust to variability parameters at the group-level: An image reconstruction investigation. Neuroimage 237, 118068 (2021). This is reference No. 68 in the manuscript.

As we mentioned in the section fNIRS preprocessing and data-analysis: ‘The sections were established via the 17 channels of each hemisphere which were grouped into front, middle and back (for a total of six regions) based on a previous co-registration of the BRIGHT fNIRS arrays onto age-appropriate templates’. The procedure mentioned by the reviewer, involving the examination of pictures showing the placement of headbands on participants, aimed to exclude infants with excessive cap displacement from further analysis.

- *The authors regress the global signal to remove systemic physiological noise. While the authors also report the changes in connectivity without global signal regression, there are some critical differences. In particular, the apparent decrease in frontal inter-hemispheric connections is not present when global signal regression is omitted, even though it is present for deoxy-Hb. The authors use connectivity results obtained after applying global signal regression for further analysis. The choice of regressing the global signal is questionable since it has been shown to introduce anti-correlations in fMRI data (Murphy et al., 2009), and fNIRS in young infants does not seem to be highly affected by physiological noise (Emberson et al., 2016). Systemic physiological noise might change at different ages, which makes its removal critical to investigate functional network development. However, global signal regression might also affect the data differently. The study would have benefited from having short separation channels to measure the systemic psychological component in the data.*

The work of Emberson et. al (2016) mentioned by the reviewer highlights indeed the challenges of removing systemic changes from the infants’ haemodynamic signal with short-channel separation (SSC). In fact, even a SSC of 1 cm detected changes in the blood in the brain, therefore by regressing this signal from the recorded one, the authors removed both systemic changes AND haemodynamic signal. This paper from Emberson et. al (2016) is taken

as a reference in the field to suggest that SSC might not be an ideal tool to remove systemic changes when collecting fNIRS data on young infants, as we did in this work.

We agree with the reviewer's observation that systemic physiological noise may vary with age and among infants. Therefore, for each infant at each age, we regressed the mean value calculated across all channels. This ensures that the regressed signal is not biased by averaged calculations at group levels.

We are aware of the criticisms directed towards global signal regression in the fMRI literature, although some other works showed anticorrelations in functional connectivity networks both with and without global signal regression (Chaia, 2012). Furthermore, Murphy himself revised his criticism on the use of global signal regression in functional connectivity analysis in one of his more recent works (Murphy et al, 2017). The fact that the decreased FC is significant in results from data pre-processed without global signal regression gives us confidence that this finding is statistically robust and not solely driven by this preprocessing choice in our pipeline.

An interesting study by Abdalmalak et al. (2022) demonstrated that failing to correct for systemic changes using any method is inappropriate when estimating FC with fNIRS, as it can lead to a high risk of elevated connectivity across the whole brain (see Figure 4 of the mentioned paper). Consequently, we strongly advocate for the implementation of global signal regression in our analysis pipeline as a fundamental step for accurate functional connectivity estimations.

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- I believe the authors bypass a fundamental point in their framing. When discussing the results, the authors compare the developmental trajectories of the infants tested in a rural area of Gambia with the trajectories reported in previous studies on infants growing in occidental high-income countries (likely in urban contexts) and attribute the differences to adverse effects (i.e., nutritional deficits). Differences in developmental trajectories might also derive from other environmental and cultural differences that do not necessarily lead to poor cognitive development.

We agree with the reviewer that other factors differing between low- and poor-resource settings might have an impact on FC trajectories. We therefore specified this in the discussion as follows: “We acknowledge that differences in FC could also be attributed to other environmental and cultural disparities between high-resource and low-resource settings, and future studies are needed to investigate this further” (line 238).

- While the study provides a solid description of the functional connectivity changes in the first two years of life at the group level, the evidence regarding the links between adverse situations, developmental trajectories, and later cognitive capacities is weaker. The authors find that early restricted growth predicts specific connectivity patterns at 24 months and that certain connectivity patterns at specific ages predict cognitive flexibility. However, the link between development trajectories (individual changes in connectivity) with growth and later cognitive capacities is missing. To address this question adequately, the study should have compared infants with different growing profiles or those who suffered or did not from undernutrition. However, as the authors discussed, they lacked statistical power.

We agree with the reviewer, and indeed we highlighted this as one of the main limitation of our work: “Even given the large sample in our study, we were underpowered to test for group comparisons between sets of infants with distinct undernutrition growth profiles, e.g., infants with early poor growth that later resolved and infants with standard growth early that had a poor growth later. We were also underpowered to test the associations between early growth and FC on clinically undernourished infants (defined as having DWLZ two standard deviations below the mean) (line 311, discussion section).

We believe this is an important point to consider for the field, as it addresses the sample size required for studies investigating brain development in clinically malnourished infants. We hope this will serve as a valuable reference for future studies in the field. For example, a new study led by Prof. Sophie Moore and other members of the BRIGHT team (INDiGO) is currently recruiting six-hundreds pregnant women with the aim of obtaining a broader distribution of infants’ growth measures (<https://www.kcl.ac.uk/research/sophie-moore-research-group>).

Reviewer #2 (Public Review):

Summary and strengths:

The article pertains to a topic of importance, specifically early life growth faltering, a marker of undernutrition, and how it influences brain functional connectivity and cognitive development. In addition, the data collection was laborious, and data preprocessing was quite rigorous to ensure data quality, utilizing cutting-edge preprocessing methods.

We thank the reviewer for highlighting the strengths of this work.

Weaknesses:

However, the subsequent analysis and explanations were not very thorough, which made some results and conclusions less convincing. For example, corrections for multiple tests need to be consistently maintained; if the results do not survive multiple corrections, they should not be discussed as significant results. Additionally, alternative plans for analysis strategies could be worth exploring, e.g., using ΔFC in addition to FC at a certain age. Lastly, some analysis plans lacked a strong theoretical foundation, such as the relationship between functional connectivity (FC) between certain ROIs and the development of cognitive flexibility.

Thus, as much as I admire the advanced analysis of connectivity that was conducted and the uniqueness of longitudinal fNIRS data from these samples (even the sheer effort to collect fNIRS longitudinally in a low-income country at such a scale!), I have reservations about the importance of this paper's contribution to the field in its present form. Major revisions are needed, in my opinion, to enhance the paper's quality.

We acknowledge the reviewer's concern regarding the reporting of results that do not survive multiple comparisons. However, considering the uniqueness of our dataset and the novelty of our work, we believe it is crucial to report all significant findings as well as hypothesis-generating findings that may not pass stringent significance thresholds. We have taken great care to transparently distinguish between results that survived multiple comparisons and those that did not in both the Results and Discussion sections, ensuring that readers are not misled. It is possible that future studies may replicate and further strengthen these associations. Therefore, by sharing these results with the research community, we provide valuable insights for future investigations.

The relationship between FC and cognitive flexibility (as well as the relationship between growth and FC) has been explored focusing on those FC that showed a significant change with age, as specified in the results sections: 'To investigate the impact of early nutritional status on FC at 24 months, we used multiple regression with the infant growth trajectory [...] and FC at 24 months [...]. To maximise power, we considered only those FC that showed a statistically significant change with age' (line 183) and 'To investigate whether FC early in life predicted cognitive flexibility at preschool age, we used multiple regression of FC across the first two years of life against later cognitive flexibility in preschoolers at three and five years. As per the analysis above, we focused on only those FC that showed a statistically significant change with age' (line 198).

We explored the possibility of investigating the relationship between changes in FC and changes in growth. However, the degrees of freedom in these analyses dropped dramatically (~25/30), thereby putting the significance and the meaning of the results at risk. We look forward to future longitudinal studies with less attrition across these time points to maintain the statistical power necessary to run such analyses.

Reviewer #3 (Public Review):

Summary:

This study aimed to investigate whether the development of functional connectivity (FC) is modulated by early physical growth and whether these might impact cognitive development in childhood. This question was investigated by studying a large group of infants (N=204) assessed in Gambia with fNIRS at 5 visits between 5 and 24 months of age. Given the complexity of data acquisition at these ages and following data processing, data could be analyzed for 53 to 97 infants per age group. FC was analyzed considering 6 ensembles of brain regions and thus 21 types of connections. Results suggested that: i) compared to previously studied groups, this group of Gambian infants have different FC trajectory, in particular with a change in frontal inter-hemispheric FC with age from positive to null values; ii) early physical growth, measured through weight-for-length z-scores from birth on, is associated with FC at 24 months. Some relationships were further observed between FC during the first two years and cognitive flexibility at 4-5 years of age, but results did not survive corrections for multiple comparisons.

Strengths:

The question investigated in this article is important for understanding the role of early growth and undernutrition on brain and behavioral development in infants and children. The longitudinal approach considered is highly relevant to investigate neurodevelopmental trajectories. Furthermore, this study targets a little-studied population from a low-/middle-income country, which was made possible by the use of fNIRS outside the lab environment. The collected dataset is thus impressive and it opens up a wide range of analytical possibilities.

We thank the reviewer for highlighting the strengths of this work.

Weaknesses:

- Analyzing such a huge amount of collected data at several ages is not an easy task to test developmental relationships between growth, FC, and behavioral capacities. In its present form, this study and the performed analyses lack clarity, unity and perhaps modeling, as it suggests that all possible associations were tested in an exploratory way without clear mechanistic hypotheses. Would it be possible to specify some hypotheses to reduce the number of tests performed? In particular, considering metrics at specific ages or changes in the metrics with age might allow us to test different hypotheses: the authors might clarify what they expect specifically for growth-FC-behaviour associations. Since some FC measures and changes might be related to one another, would it be reasonable to consider a dimensionality reduction approach (e.g., ICA) to select a few components for further correlation analyses?

We confirm that this work was motivated by a compelling theoretical question: whether neural mechanisms, specifically FC, can be influenced by early adversity, such as growth, and subsequently impact cognitive outcomes, such as cognitive flexibility. This aligns with the overarching goal of the BRIGHT project, established in 2015 (Lloyd-Fox, 2023). We believe this was evident throughout the manuscript in several instances, for example:

- “The goal of the study was to investigate early physical growth in infancy, developmental trajectories of brain FC across the first two years of life, and cognitive outcome at school age in a longitudinal cohort of infants and children from rural Gambia, an environment with high rates of maternal and child undernutrition. Specifically, we aimed to: (i) investigate whether differences in physical growth through the first two years of life are related to FC at 24 months, and (ii) investigate if trajectories of early FC have an impact on cognitive outcome at pre-school age in these children.” (page 4, introduction)

- “This study investigated how early adversity via undernutrition drives longitudinal changes in brain functional connectivity at five time points throughout the first two years of life and how these developmental trajectories are associated with cognitive flexibility at preschool age.” (page 6, discussion)

- We had a clear hypothesis regarding short-range connectivity decreasing with age and long-range connectivity increasing with age, as stated at the end of the introduction: We hypothesized that (i) long-range FC would increase and short-range FC would decrease throughout the first two years of life” (page 4, line 147). However, we were not able to formulate clear hypotheses about the localization of these connections due to the scarcity of previous studies conducted within this age range, particularly in low-resource settings. The ROI approach for analysis was chosen to mitigate this challenge by reducing the number of comparisons while still enabling us to estimate the developmental trajectories of all the connections from which we acquired data.

Regarding the use of dimensionality reduction approach, we have not considered the use of ICA in our analysis. These methods require selecting a fixed number of components to remove from all participants. However, due to the high variability of infant fNIRS data across the five timepoints, we considered it untenable to precisely determine the number of components to remove at the group level. Such a procedure carries the risk of over-cleaning the data for some participants while leaving noise in for others (Di Lorenzo, 2019). We also felt that using PCA in this initial study would be beyond the scope of the brain-region-specific hypotheses and would be more appropriate in a follow-up analysis of these important data.

References:

Lloyd-Fox, S., McCann, S., Milosavljevic, B., Katus, L., Blasi, A., Bulgarelli, C., Crespo-Llado, M., Ghillia, G., Fadera, T., Mbye, E., Mason, L., Njai, F., Njie, O., Perapoch-Amado, M., Rozhko, M., Sosseh, F., Saidykhan, M., Touray, E., Moore, S. E., ... Team, and the B. S. (2023). The Brain Imaging for Global Health (BRIGHT) Study: Cohort Study Protocol. *Gates Open Research*, 7(126).

Di Lorenzo, R., Pirazzoli, L., Blasi, A., Bulgarelli, C., Hakuno, Y., Minagawa, Y., & Brigadoi, S. (2019). Recommendations for motion correction of infant fNIRS data applicable to multiple data sets and acquisition systems. *NeuroImage*, 200(April), 511–527.

- It seems that neurodevelopmental trajectories over the whole period (5-24 months) are little investigated, and considering more robust statistical analyses would be an important aspect to strengthen the results. The discussion mentions the potential use of structural equation modelling analyses, which would be a relevant way to better describe such complex data.

We appreciate the complexity of the dataset we are working with, which includes multiple measures and time points. Currently, our focus within the outputs from the BRIGHT project is on examining the relationship between selected measures. While this may not involve statistically advanced modelling at the moment, it is worth noting that most of the results presented in this work have survived correction for multiple comparisons, indicating their statistical robustness. We believe that more advanced statistical analyses are beyond the scope of this rich initial study. In the next phase of the project, known as BRIGHT IMPACT, our team is collaborating with statisticians and experts in statistical modelling to apply more sophisticated and advanced statistical techniques to the data.

- Given the number of analyses performed, only describing results that survive correction for multiple comparisons is required. Unifying the correction approach (FDR / Bonferroni) is also recommended. For the association between cognitive flexibility and FC, results are not significant, and one might wonder why FC at specific ages was considered rather than the change in FC with age. One of the relevant questions of such a study would be whether early growth and later cognitive flexibility are related through FC development, but testing this would require a mediation analysis that was not performed.

We acknowledge the reviewer's concern regarding the reporting of results that do not survive multiple comparisons. However, considering the uniqueness of our dataset and the novelty of our work, we believe it is crucial to report all significant findings. We have taken great care to transparently distinguish between results that survived multiple comparisons and those that did not in both the Results and Discussion sections, ensuring that readers are not misled. It is possible that future studies may replicate and further strengthen these associations. Therefore, by sharing these results with the research community, we provide valuable insights for future investigations.

We did not perform a mediation analysis as i) Δ WLZ between birth and the subsequent time points positively predicted frontal interhemispheric FC at 24 months, ii) frontal interhemispheric FC at 18 months (and right fronto-posterior connectivity at 24 months) predicted cognitive flexibility at preschool age. Considering that the frontal interhemispheric FC at 24 months that was positively predicted by growth, did not significantly predict cognitive outcome at preschool age, we did not perform mediation models.

The reviewer raised concerns about using different methods to correct for multiple comparisons throughout the work. Results showing changes in FC with age were Bonferroni corrected, while we used FDR correction for the regression analyses investigating the

relationship between growth and FC, as well as FC and cognitive flexibility. Both methods have good control over Type I errors (false positives), but Bonferroni is very conservative, increasing the likelihood of Type II errors (false negatives). We considered Bonferroni an appropriate method for correcting results showing changes in FC with age, where we had a large sample with strong statistical power (i.e. linear mixed models with 132 participants who had at least 250 seconds of good data for 2 out of 5 visits). However, Bonferroni was too conservative for the regression analyses, with N between 57 and 78) (Acharya, 2014; Félix & Menezes, 2018; Narkevich et al., 2020; Narum, 2006; Olejnik et al., 1997).

References:

- Acharya, A. (2014). A Complete Review of Controlling the FDR in a Multiple Comparison Problem Framework--The Benjamini-Hochberg Algorithm. ArXiv Preprint ArXiv:1406.7117.
- Félix, V. B., & Menezes, A. F. B. (2018). Comparisons of ten corrections methods for t-test in multiple comparisons via Monte Carlo study. *Electronic Journal of Applied Statistical Analysis*, 11(1), 74–91.
- Narkevich, A. N., Vinogradov, K. A., & Grjibovski, A. M. (2020). Multiple comparisons in biomedical research: the problem and its solutions. *Ekologiya Cheloveka (Human Ecology)*, 27(10), 55–64.
- Narum, S. R. (2006). Beyond Bonferroni: less conservative analyses for conservation genetics. *Conservation Genetics*, 7, 783–787.
- Olejnik, S., Li, J., Supattathum, S., & Huberty, C. J. (1997). Multiple testing and statistical power with modified Bonferroni procedures. *Journal of Educational and Behavioral Statistics*, 22(4), 389–406.

- Growth is measured at different ages through different metrics. Justifying the use of weight-for-length z-scores would be welcome since weight-for-age z-scores might be a better marker of growth and possible undernutrition (this impacting potentially both weight and length). Showing the distributions of these z-scores at different ages would allow the reader to estimate the growth variability across infants.

We consistently used WLZ as the metric to measure growth throughout. Our analysis investigating the relationship between WLZ and growth included HCZ at 7/14 days to correct for head size at birth. When selecting the best growth measure for this paper, we opted for WLZ over WAZ, given extant evidence that infants in our sample are smaller and shorter compared to the reference WHO standard for the same age group (Nabwera et al., 2017). Therefore, using WLZ allows us to adjust each infant's weight for its own length.

References:

- Nabwera, H. M., Fulford, A. J., Moore, S. E., & Prentice, A. M. (2017). Growth faltering in rural Gambian children after four decades of interventions: a retrospective cohort study. *The Lancet Global Health*, 5(2), e208–e216.

- Regarding FC, clarifications about the long-range vs short-range connections would be welcome, as well as drawing a summary of what is expected in terms of FC "typical" trajectory, for the different brain regions and connections, as a marker of typical development. For instance, the authors suggest that an increase in long-range connectivity vs a decrease in short-range is expected based on previous fNIRS studies. However anatomical studies of white matter growth and maturation would suggest the reverse pattern (short-range connections developing mostly after birth, contrarily to long-range connections prenatally).

We expected an increase in long-range functional connectivity with age, as discussed in the introduction:

- “Based on data from fMRI, current models hypothesize that FC patterns mature throughout early development (23–27), where in typically developing brains, adult-like networks emerge over the first years of life as long-range functional connections between pre-frontal, parietal, temporal, and occipital regions become stronger and more selective (28–31). This maturation in FC has been shown to be related to the cascading maturation of myelination and synaptogenesis (32, 33) - fundamental processes for healthy brain development (34)” (line 93, page 3, introduction);

- “Importantly, normative developmental patterns may be disrupted and even reversed in clinical conditions that impact development; e.g., increased short-range and reduced long-range FC have been observed in preterm infants (36) and in children with autism spectrum disorder (37, 38)” (line 103, page 3, introduction);

- “We hypothesized that (i) long-range FC would increase and short-range FC would decrease throughout the first two years of life” (line 147, page 4, introduction).

Since inferences about FC patterns recorded with fNIRS are highly limited by the number and locations of the optodes, it is challenging to make strong inferences about specific brain regions. Moreover, infant FC fNIRS studies are still limited, which is why we focused our inferences on long-range versus short-range connectivity, without specifically pinpointing particular brain regions.

Additionally, we were unable to locate the works mentioned by the reviewer regarding an increase in short-range white matter connectivity immediately after birth. On the contrary, we found several studies documenting an increase in white-matter long-range connectivity after birth, which is consistent with the hypothesised increase in FC long-range connectivity, such as:

Yap, P. T., Fan, Y., Chen, Y., Gilmore, J. H., Lin, W., & Shen, D. (2011). Development trends of white matter connectivity in the first years of life. *PloS one*, 6(9), e24678.

Dubois, J., Dehaene-Lambertz, G., Kulikova, S., Poupon, C., Hüppi, P. S., & Hertz-Pannier, L. (2014). The early development of brain white matter: a review of imaging studies in fetuses, newborns and infants. *Neuroscience*, 276, 48-71.

Stephens, R. L., Langworthy, B. W., Short, S. J., Girault, J. B., Styner, M. A., & Gilmore, J. H. (2020). White matter development from birth to 6 years of age: a longitudinal study. *Cerebral Cortex*, 30(12), 6152-6168.

Hagmann, P., Sporns, O., Madan, N., Cammoun, L., Pienaar, R., Wedeen, V. J., ... & Grant, P. E. (2010). White matter maturation reshapes structural connectivity in the late developing human brain. *Proceedings of the National Academy of Sciences*, 107(44), 19067-19072.

Collin G, van den Heuvel MP. The ontogeny of the human connectome: development and dynamic changes of brain connectivity across the life span. *Neuroscientist*. 2013 Dec;19(6):616-28. doi: 10.1177/1073858413503712.

The authors test associations between FC and growth, but making sense of such modulation results is difficult without a clearer view of developmental changes per se (e.g., what does an early negative FC mean? Is it an increase in FC when the value gets close to 0? In particular, at 24m, it seems that most FC values are not significantly different from 0, Figure 2B). Observing positive vs negative association effects depending on age is quite puzzling. It is also questionable, for some correlation analyses with

cognitive flexibility, to focus on FC that changes with age but to consider FC at a given age.

We thank the reviewer for bringing up this important point and understand that it requires some additional consideration. The negative FC values decreasing with age indicate that these regions go from being anti-correlated to becoming increasingly correlated. Hence, FC of these ROIs increased with age. The trajectory seems to suggest that this will keep increasing with age but of course further data need to be collected to assess this.

Unfortunately, when considering ΔFC to predict cognitive flexibility, the numbers of participants dropped significantly, with $N \sim 15/20$ infants per group of preschoolers, making it very challenging to interpret the results with meaningful statistical power.

- The manuscript uses inappropriate terms "to predict", "prediction" whereas the conducted analyses are not prediction analyses but correlational.

We thank the reviewer for giving us to opportunity to thoroughly revise the manuscript about this matter. In this work, we had clear hypotheses regarding which variables predicted which certain measures (such as growth predicting FC and FC predicting cognitive outcomes). Therefore, we performed regression analyses rather than correlational analyses to investigate these associations. Hence, we believe that using the term 'predict and 'prediction' is appropriate

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) In the introduction and discussion, the authors talk about the link between developmental trajectories and cognitive capacities, and undernutrition. However, they did not compare developmental trajectories but connectivity patterns at different ages with ΔWLZ and cognitive flexibility. I recommend that the authors rephrase the introduction and discussion.

We thank the reviewer for pointing out places requiring better clarity in the text. We made edits through the introduction to better match our investigations. In particular we changed:

- 'our understanding of the relationships between early undernutrition, developmental trajectories of brain connectivity, and later cognitive outcomes is still very limited,' to, 'our understanding of the relationships between early undernutrition, brain connectivity, and later cognitive outcomes is still very limited' (line 89, introduction);
- '(ii) investigate if trajectories of early FC have an impact on cognitive outcome at pre-school age in these children,' to, '(ii) investigate if early FC has an impact on cognitive outcome at pre-school age in these children' (line 137, introduction);
- 'This study investigated how early adversity via undernutrition drives longitudinal changes in brain functional connectivity at five time points throughout the first two years of life and how these developmental trajectories are associated with cognitive flexibility at preschool age,' to, 'This study investigated how early adversity via undernutrition drives brain functional connectivity throughout the first two years of life and how these early functional connections are associated with cognitive flexibility at preschool age' (line 215, discussion).

(2) Considering most research is done in occidental high-income countries, and this work is one of the few presenting research in another context, I think the authors should discuss in the manuscript that differences with previous studies might also be due to environmental and cultural differences. Since the study lacks the statistical power to

perform a statistical analysis that directly establishes a link between developmental trajectories and restricted growth and cognitive flexibility, the authors cannot disentangle which differences are related to undernutrition and which might result from growing up in a different environment. I recommend that the authors avoid phrases like (lines 57-58): "We observed that early physical growth before the fifth month of life drove optimal developmental trajectories of FC..." or (lines 223-224) "...our cohort of Gambian infants exhibit atypical developmental trajectories of functional connectivity..."

We thank the reviewer for this observation, and we agree with the reviewer that other factors differing between low- and poor-resource settings might have an impact on FC trajectories. We therefore specified this in the discussion as follows: "We acknowledge that differences in FC could also be attributed to other environmental and cultural disparities between high-resource and low-resource settings, and future studies are needed to explore this further" (line 238). We revised the whole manuscript to reflect similar statements.

(3) To better interpret the results, it would be interesting to know if poor early growth predicts late cognitive flexibility in the tested sample and if the Δ WLZ distributions differ compared to a population in a high-income country where undernutrition is less frequent.

We explored the relationship between changes in growth and cognitive flexibility in the two preschooler group, but there were no significant associations.

Mean and SD values of WLZ are reported in Table 3. The values at every age are negative, indicating that the infants' weight-for-length is below the expected norm at all ages. To our knowledge, no other studies have assessed changes in growth in an infant sample with similar closely spaced age time points in high-income countries, making comparisons on growth changes challenging.

(4) It is unclear why WLZ at birth and HCZ at 7-14 days are included in the models. I imagine this is to ensure that differences are not due to growing restrictions before birth. It would be nice if the authors could explain this.

As the reviewer pointed out, HCZ at 7-14 days was included to ensure associations between growth and FC are not due to physical differences at birth. This case be considered as a 'baseline' measure for cerebral development, in the same way that WLZ at birth was used as a baseline for physical development. Therefore, we can more confidently assume that the associations between growth and FC were specific to the impact of change in WLZ postnatally and not confounded by the size or maturity of the infant at birth. We specified this in the manuscript as follows: "These analyses were adjusted by WLZ at birth and HCZ at 7/14 days, to more confidently assume that the associations between growth and FC were specific to the impact of change in WLZ postnatally and not confounded by the size or maturity of the infant at birth" (line 520, statistical analysis section in the method section).

(5) Right frontal-posterior connections at 24 months negatively correlate with Δ WLZ. Thus, restricted growth results in stronger frontal-posterior connections at 24 months. However, the same connections at 24 months positively correlate with cognitive flexibility (stronger connections predict better cognitive flexibility). Do the authors have any interpretation of this? How could this relate to previous findings of the authors (Bulgarelli et al. 2020), showing first an increase and then a decrease in functional connectivity between frontal and parietal regions?

We acknowledge that interpreting the negative relationship between changes in growth and fronto-posterior FC at 24 months, alongside the positive association between the same

connection and later cognitive flexibility, is challenging. We refrain from relating these findings to those published by Bulgarelli in 2020 due to differences in optode locations and because in that work the decrease in fronto-posterior FC was observed after 24 months (up to 36 months), whereas the endpoint in this study is right at 24 months.

(6) With the growth of the head, the frontal channels move to more temporal areas, right? Could this determine the decrease in frontal inter-hemisphere connections?

As shown in Nabwera (2017) head size does not increase that much in Gambian infants, or at least as expected by the WHO standard measures. We have added HCZ mean and SD values per age in Table 3.

Minor points

- HCZ is used in line 184 but not defined.

We thank the reviewer for spotting this, we have now specified HCZ at line 184 as follows: 'head-circumference z-score (HCZ)'.

- Table SI2: NIRS not undertaken = the participant was assessed but did want or could not perform... I imagine there is a missing "not".

We thank the reviewer for spotting this, we have now modified the legend of Table SI2 as follows: 'the participant was assessed but did not want or could not perform the NIRS assessments.'

- The authors should explain what weight-for-length is for those who are not familiar with it.

We have added an explanation of weight-for-length in the experimental design section, line 339 as follows: 'We then tested for relationships between brain FC at age 24 months with measures of early growth, as indexed by changes in weight-for-length z-scores (reflecting body weight in proportion to attained growth in length) at one month of age, and at each of the four subsequent visits (details provided below).'

Reviewer #2 (Recommendations For The Authors):

(1) I am confused about the authors' interpretation that left and right front-middle and right front-back FC increased with age. It appears in Figure 2 that the negative FC among these ROIs should actually decrease with age. This means that as individuals grow older, the FC values between these regions and zero diminished, albeit starting with negative FC (anticorrelation values) in younger age groups.

Yes, the reviewer is correct. The negative values of the left and right front-middle and right front-back FC decreasing with age indicate that these regions go from being anti-correlated to becoming increasingly correlated. Hence, FC of these ROIs increased with age.

(2) Are these negative values mentioned above at 24 months still negative? Have t-tests been run to examine the differences from zero?

As suggested, we performed t-tests against zero for the mentioned FC at 24 months, and only the left and right fronto-middle FC are significantly different than zero (left fronto-middle FC: $t(94) = 1.8$, $p = 0.036$; right fronto-middle FC $t(94) = 2.7$, $p = 0.003$).

(3) With so many correlation analyses, have multiple comparisons been consistently controlled for? While I assume this was done according to the Methods section, could the authors clarify whether FDR adjustment was applied to all the p-values at once or to a group of p-values each time? I found the following way of reporting FDR-adjusted p-values quite informative, such as PFDR, 24 pairs of ROIs < 0.05.

We thank the reviewer for this insightful comment. P-values of regression analyses were FDR corrected per connection investigated, i.e. 21 possible Δ WLZ values per connection. We have specified this in the method section as follows: “To ensure statistical reliability, results from the regression analyses on each FC were corrected for multiple comparisons using false discovery rate (FDR)(Benjamini & Hochberg, 1995) per each connection investigated, i.e. 21 possible Δ WLZ values per each connection,” (page 12, Statistical Analyses section).

(4) Can early growth trajectories predict changes in FC? Why not use Δ WLZ to predict Δ FC?

Unfortunately, when considering Δ WLZ to predict Δ FC, the numbers of participants dropped significantly, with $N \sim 30$ infants, making it very challenging to interpret the results. We believe this emphasizes the importance of recruiting large samples when conducting longitudinal studies involving infants and employing multiple measures.

(5) I might have missed the rationale, but why weren't the growth changes after 5 months studied?

Δ WLZ between all time points were assessed as predictors of FC at 24 months. We have specified this at line 183 as follows: ‘we used multiple regression with the infant growth trajectory (delta weight for length z-score between all time points, DWLZ) and FC at 24 months’. As indicated in Table 2 and 3 the associations between Δ WLZ at all time points and FC at 24 months were tested, but only those with DWLZ calculated between birth and 1 month and the subsequent time points were significant. DWLZ between 5 months and the subsequent time points, DWLZ between 8 months and the subsequent time points, DWLZ between 12 months and the subsequent time points, DWLZ between 18 months and the subsequent time points did not significantly predict FC at 24 months. These are highlighted in Table 2 and Figure 3 in blue and marked as NS (non-significant).

(6) Once more, the advantage of longitudinal data is that it allows us to tap into developmental changes. Analyzing and predicting cognitive development based solely on FC values at a single age stage (i.e., 24 months) would overlook the benefits of a longitudinal design, which is regrettable. I suggest that the authors attempt to use Δ FC for predictions and observe the outcomes.

As mentioned to point (4) raised by the reviewer, unfortunately, when considering Δ WLZ to predict Δ FC, the numbers of participants dropped significantly, with $N \sim 30$ infants, making it very challenging to interpret the results. We believe this emphasizes the importance of recruiting large samples when conducting longitudinal studies involving infants and employing various measures.

(7) In the section "Early FC predicts cognitive flexibility at preschool age", the authors pointed out that "...none of these survived FDR correction for multiple comparisons." However, the paper discussed the association between FC at 24 months of age and cognitive flexibility, as it was supported by the statistical analysis in the following sections. If FDR correction cannot be satisfied, I would rephrase the

implication/conclusion of the results to suggest that early FC does not predict cognitive flexibility at preschool age.

We acknowledge the reviewer's concern regarding the reporting of results that do not survive multiple comparisons. However, considering the uniqueness of our dataset and the novelty of our work, we believe it is crucial to report all significant findings, even those not passing multiple comparisons corrections, as they may motivate hypothesis-generation for future studies. We have taken great care to transparently distinguish between results that survived multiple comparisons and those that did not in both the Results and Discussion sections, ensuring that readers are not misled. It is possible that future studies may replicate and further support these associations. Therefore, by sharing these results with the research community, we provide valuable insights for future investigations.

Following the reviewer's suggestion, we specified that results from regression analysis are significant but they did not survive multiple comparisons in the discussion as follows: 'While our results are consistent with previous studies, we acknowledge that the significant association between early FC and later cognitive flexibility does not withstand multiple comparisons. Therefore, we encourage future studies that may replicate these findings with a larger sample. (line 290, discussion section).

(8) Have the authors assessed the impact of growth trajectories on cognitive flexibility?

We explored the relationship between changes in growth and cognitive flexibility in the two preschooler groups, but there were no significant associations.

(9) Are there no other cognitive or behavioural measures available? Cognitive flexibility is just one domain of cognitive development, and would the impact of undernutrition on cognitive development be domain-specific? There is a lack of theoretical support here. Why choose cognitive flexibility, and should the impact of undernutrition be domain-specific or domain-general?

We agree with the reviewer that in this work, we chose to focus on one specific cognitive outcome. While this does not imply that the impact of undernutrition is domain-specific, cognitive flexibility, being a core executive function, has been extensively studied in terms of its neural underpinnings using other neuroimaging modalities, especially fMRI (for example see Dajani, 2015; Uddin, 2021).

Moreover, other studies looking at the effect of adversity on cognitive outcomes focus on specific cognitive skills, such as working memory (Roberts, 2017), reading and arithmetic skills (Soni, 2021).

We did assess infants also with Mullen Scales of Early Learning (MSEL), although the cognitive flexibility task within the Early Years Toolbox has been specifically designed for preschoolers (Howard, 2015), and this set of tasks has recently been validated in our team in The Gambia (Milosavljevic, 2023). Future works from the BRIGHT team will investigate performance at the MSEL in relation to other variable of the project.

References:

- D. R. Dajani, L. Q. Uddin, Demystifying cognitive flexibility: Implications for clinical and developmental neuroscience. *Trends Neurosci.* 38, 571–578 (2015).
- L. Q. Uddin, Cognitive and behavioural flexibility: neural mechanisms and clinical considerations. *Nat. Rev. Neurosci.* 22, 167–179 (2021).

Roberts, S. B., Franceschini, M. A., Krauss, A., Lin, P. Y., de Sa, A. B., C , R., ... & Muentener, P. (2017). A pilot randomized controlled trial of a new supplementary food designed to enhance cognitive performance during prevention and treatment of malnutrition in childhood. *Current developments in nutrition*, 1(11), e000885.

Soni, A., Fahey, N., Bhutta, Z. A., Li, W., Frazier, J. A., Moore Simas, T., ... & Allison, J. J. (2021). Early childhood undernutrition, preadolescent physical growth, and cognitive achievement in India: A population-based cohort study. *PLoS Medicine*, 18(10), e1003838.

Howard, S. J., & Melhuish, E. (2015). An Early Years Toolbox (EYT) for assessing early executive function, language, self-regulation, and social development: Validity, reliability, and preliminary norms. *Journal of Psychoeducational Assessment*, 35(3), 255-275.

Milosavljevic, B., Cook, C. J., Fadera, T., Ghillia, G., Howard, S. J., Makaula, H., ... & Lloyd-Fox, S. (2023). Executive functioning skills and their environmental predictors among pre-school aged children in South Africa and The Gambia. *Developmental Science*, e13407.

(10) I would review more previous fNIRS studies on infants if they exist (e.g., the work by S Lloyd-Fox, L Emberson, and others). These studies can help identify brain ROIs likely linked to undernutrition and cognitive flexibility. The current analysis methods lean towards exploratory research. This makes the paper more of a proof-of-concept report rather than a strongly theoretically-driven study.

We thank the reviewer for this important point. While we have reviewed existing fNIRS infant studies, there are no extant works that showed whether specific brain regions are related undernutrition. However, several fMRI studies assessed regions that do support cognitive flexibility, and we mentioned these in the manuscript (for example see Dajani, 2015; Uddin, 2021).

Other than the BRIGHT project, we are aware of two other projects that assessed the effect of undernutrition on brain development, assessing cognitive outcomes in poor-resource settings:

- the BEAN project in Bangladesh in which fNIRS data were recorded from the bilateral temporal cortex (i.e. Pirazzoli, 2022);
- a project in India in which fNIRS data were recorded from frontal, temporal and parietal cortex bilaterally (i.e. Delgado Reyes, 2020)

The brain regions recorded in these studies largely overlap with the brain regions we recorded from in this study.

Another aspect to consider is that infants underwent several fNIRS tasks as part of the BRIGHT project, focusing on social processing, deferred imitation, and habituation responses. Therefore, brain regions for data acquisition were chosen to maximize the likelihood of recording meaningful data for all tasks (Lloyd-Fox, 2023). To clarify the text, we specified this information in the methods section (line 383).

References:

D. R. Dajani, L. Q. Uddin, Demystifying cognitive flexibility: Implications for clinical and developmental neuroscience. *Trends Neurosci.* 38, 571–578 (2015).

Pirazzoli, L., Sullivan, E., Xie, W., Richards, J. E., Bulgarelli, C., Lloyd-Fox, S., ... & Nelson III, C. A. (2022). Association of psychosocial adversity and social information processing in children raised in a low-resource setting: an fNIRS study. *Developmental Cognitive Neuroscience*, 56, 101125.

Delgado Reyes, L., Wijeakumar, S., Magnotta, V. A., Forbes, S. H., & Spencer, J. P. (2020). The functional brain networks that underlie visual working memory in the first two years of life. *NeuroImage*, 219, Article 116971.

Lloyd-Fox, S., McCann, S., Milosavljevic, B., Katus, L., Blasi, A., Bulgarelli, C., Crespo-Llado, M., Ghillia, G., Fadera, T., Mbye, E., Mason, L., Njai, F., Njie, O., Perapoch-Amado, M., Rozhko, M., Sosseh, F., Saidykhan, M., Touray, E., Moore, S. E., ... Team, and the B. S. (2023). The Brain Imaging for Global Health (BRIGHT) Study: Cohort Study Protocol. *Gates Open Research*, 7(126).

(11) Last but not least, in the paper, the authors mentioned that fNIRS offers better spatial resolution and anatomical specificity compared to EEG, thereby providing more precise and reliable localization of brain networks. While I partially agree with this perspective, it remains to be explored whether the current fNIRS analysis strategies indeed yield higher spatial resolution. It is hoped that the authors will delve deeper into this discussion in the paper.

The brain regions of focus were selected based on coregistration work previously conducted at each time point on the array used in this project (Collins-Jones, 2019). We deliberately avoided making claims about small brain regions, considering that head size might increase slightly less with age in The Gambia compared to Western countries (Nabwera, 2017). However, we maintain that the conclusions drawn in this study offer higher brain-region specificity than could have been identified with current common EEG methods alone.

References:

L. H. Collins-Jones, et al., Longitudinal infant fNIRS channel-space analyses are robust to variability parameters at the group-level: An image reconstruction investigation. *Neuroimage* 237, 118068 (2021).

Nabwera, H. M., Fulford, A. J., Moore, S. E., & Prentice, A. M. (2017). Growth faltering in rural Gambian children after four decades of interventions: a retrospective cohort study. *The Lancet Global Health*, 5(2), e208–e216.

Reviewer #3 (Recommendations For The Authors):

Introduction

- Among important developmental mechanisms to mention are the development of exuberant connections and the further selection/stabilization of the relevant ones according to environmental stimulation, vs the pruning of others.

We agree with the reviewer that the development of exuberant connections and subsequent pruning is a universal process of paramount importance during the first years of life. However, after revising our introduction, given the word limit of the journal, we maintained focus on neurodevelopment and early adversity.

Results

- Adding a few more information on the 6 sections and 21 connections would be welcome. In particular for within-section FC: how was this computed?

The 6 sections were created based on the co-registration of the array used in this study at each age in a previous published work L. H. Collins-Jones, et al., Longitudinal infant fNIRS channel-space analyses are robust to variability parameters at the group-level: An image

reconstruction investigation. *Neuroimage* 237, 118068 (2021). This is reference No. 68 in the manuscript.

As we mentioned in the section fNIRS preprocessing and data-analysis: ‘The sections were established via the 17 channels of each hemisphere which were grouped into front, middle and back (for a total of six regions) based on a previous co-registration of the BRIGHT fNIRS arrays onto age-appropriate templates’.

The 21 connections were defined as all the possible links between the 6 regions, specifically: the interhemispheric homotopic connections (in orange in Figure SI1), which connect the same regions between hemispheres (i.e., front left with front right); the intrahemispheric connections (in green in Figure SI1), which correlate channels belonging to the same region; the fronto-posterior connections (in blue in Figure SI1), which link front and middle, middle and back, and front and back regions of the same hemisphere; and the crossing interhemispheric connections (non-homotopic interhemispheric, in yellow in Figure SI1), which link the front, middle, and back areas between left and right hemispheres. We added these specifications also in the legend of Figure SI1 for clarity.

- The denomination intrahemispheric vs fronto-posterior vs crossed connections is not clear. Maybe prefer intra-hemispheric vs inter-hemispheric homotopic vs inter-hemispheric non-homotopic (also in Figure SI1).

We appreciate the reviewer's suggestion regarding terminology. However, we believe that the term 'inter-hemispheric non-homotopic' could potentially refer to both connections within the same brain hemisphere from front to back and connections crossing between hemispheres, leading to increased confusion. Therefore, we have chosen not to include the term 'non-homotopic' and instead added 'homotopic' to 'interhemispheric' throughout the manuscript to emphasize that these functional connections occur between corresponding regions of the two hemispheres.

- with time -> with age.

We replaced “with time” with “with age” as suggested through the manuscript.

- The description of both HbO2 and HHb results overloads the main text: would it be relevant to present one of the two in Supplementary Information if the results are coherent?

We understand the reviewer's concern regarding overloading the results section with reporting both chromophores. However, reporting results for both HbO and HHb is considered a crucial step for publications in the fNIRS field, as emphasized in recent formal guidance (Yücel et al., 2020). One of the strengths of fNIRS compared to fMRI is its ability to record from both chromophores, enabling a more precise characterization of brain activations and oscillations. Moreover, in FC studies like this one, ensuring that HbO and HHb results overlap is an important check that increases confidence in interpreting the findings.

References:

Yücel, M. A., von Lüthmann, A., Scholkmann, F., Gervain, J., Dan, I., Ayaz, H., Boas, D., Cooper, R. J., Culver, J., Elwell, C. E., Eggebrecht, A., Franceschini, M. A., Grova, C., Homae, F., Lesage, F., Obrig, H., Tachtsidis, I., Tak, S., Tong, Y., ... Wolf, M. (2020). Best Practices for fNIRS publications. *Neurophotronics*, 1–34. <https://doi.org/10.1117/1.NPh.8.1.012101>

- HCZ is not defined when first used.

We thank the reviewer for spotting this, we have now specified HCZ at line 184 as follows: 'head-circumference z-score (HCZ)'.

| - *Choosing the analyzed measures to "maximize power" could be criticised.*

We appreciate the reviewer's concern. However, correlating all the FC values with all changes in growth would have raised an important issue for multiple comparisons. We therefore we made a priori decision to focus on investigating the relationship between changes in growth and those FC that showed a significant change with age, considering these as the most interesting ones from a developmental perspective in our sample.

| *Discussion*

| - *I would recommend using the same order to synthesize results and further discuss them.*

We agree with the reviewer that the suggested structure is optimal for a clear discussion section. We have indeed followed it, with each paragraph covering specific aspects:

- Recap of the study aims
- Results summary and discussion of developmental changes
- Results summary and discussion of the relationship between changes in growth and FC
- Results summary and discussion of the relationship between FC and cognitive flexibility
- Limitations
- Conclusion

Given the numerous results presented in this paper, we believe that readers will better digest them by first reading a summary of the results followed by their interpretations, rather than condensing all the interpretations together.

| - *Highlighting how "atypical" developmental trajectories are in Gambian infants would be welcome in the Results section. Other interpretations can be found than "The observed decrease in frontal inter-hemispheric FC with increasing age may be due to the exposure to early life undernutrition adversity".*

We agree with the reviewer that other factors that differ between low- and high-resource settings might have an impact on FC trajectories. We therefore specified this in the discussion as follows: "We acknowledge that differences in FC could also be attributed to other environmental and cultural disparities between high-resource and low-resource settings, and future studies are needed to further investigate cultural, environmental, and genetic effects on brain FC" (line 238).

| - *Focusing on FC at 24m for the relationship with growth is questionable.*

Correlating the FC values at 5 time points with all changes in growth would have raised an important issue for multiple comparisons. We therefore we made a decision a priori to focus on investigating the relationship between changes in growth and FC at 24 months as our final time point of data collection. We added this information in the methods section as follows: "To investigate the impact of undernutrition on FC development, we used DWLZ as independent variables in regression analyses on HbO2 (as the chromophore with the highest

signal-to-noise ratio) FC at 24 months, our final time point of data collection” (line 517, method section).

- There is too much emphasis on the correlation between FC and cognitive flexibility, whereas results are not significant after correction for multiple comparisons.

Following the reviewer’s suggestion, we specified that results from regression analysis are significant but they did not survive multiple comparisons in the discussion as follows: While our results are consistent with previous studies, we acknowledge that the significant association between early FC and later cognitive flexibility does not withstand multiple comparisons. Therefore, we encourage future studies that may replicate these findings with a larger sample. (line 290, discussion section).

Methods

- I would recommend detailing how z-scores were computed in the paragraph "Anthropometric measures".

We specified how z-scores were computed in the statistical analysis section as follows: “Anthropometric measures were converted to age and sex adjusted z-scores that are based on World Health Organization Child Growth Standards (93). Weight-for-Length (WLZ) and Head Circumference (HCZ) z-scores were computed” (line 509, method section). As transforming data is the first step of statistical analysis and is not directly related to data collection, we believe it is more appropriate to retain this description in the statistical analysis section.

- FC computation: the mention of "correlating the first and the last 250s" is not clear.

We specified this more clearly in the text as follows: We found that correlating the first and the last 250 seconds of valid data after pre-processing provided the highest percentage of infants with strong correlation between the first and the last portion of data (line 467).

- The manuscript mentions "age 3 years" for the younger preschoolers but ~48months rather corresponds to 4 years.

We revised the entire manuscript and the supplementary materials, but we could not find any instance in which preschoolers are referred with age in months rather than in years.

- Specify the number of children evaluated at 4 and 5 years. Is the test of cognitive flexibility normalized for age? If not, how were the 2 groups considered in the analyses? (age as a confounding factor).

We have added the number of children in the two preschooler groups as follows: younger preschoolers (age mean $\pm$ SD=47.96 $\pm$ 2.77 months, N=77) and older preschoolers (age mean $\pm$ SD=57.58 $\pm$ 2.11 months, N=84). (line 484).

The cognitive flexibility test was not normalized for age, as this task was specifically developed for preschoolers (Howard, 2015). As mentioned in ‘Cognitive flexibility at preschool age’ of the methods section, “data were collected in two ranges of preschool ages”, which guided our decision to perform regression analysis on the impact of FC on cognitive flexibility separately within these two age groups, rather than treating them as a single group of preschoolers.

References:

Howard, S. J., & Melhuish, E. (2015). An Early Years Toolbox (EYT) for assessing early executive function, language, self-regulation, and social development: Validity, reliability, and preliminary norms. *Journal of Psychoeducational Assessment*, 35(3), 255-275.

Figures and Tables

- Table 1 could highlight the significant results. It is not clear what the "baseline" results correspond to.

We have marked in bold the results that are statistically significant in Table 1. In the linear mixed model we performed, the first time point (i.e. 5 months) is chosen as 'baseline', i.e. the reference against which the other time points are compared to, and its statistical values refer to its significance against 0 (as it has been performed in Bulgarelli 2020).

- Figures 2 B and C seem redundant? What is SE vs SD?

We believe that both figures 2B and 2C are useful for the readers. While the first one shows the mean FC values at the group level, the second one highlights the individual variability of FC values (typical of infant neuroimaging data), which also why it is interesting to relate these measures to other variables of our dataset (i.e. growth and cognitive flexibility). Figure 2C also reports mean FC values per age, but these might be less visible considering that also one dot per infant is also plotted.

SE stands for standard error, and in the legend of the figure we specified this as follows: 'Mean and standard error of the mean (SE)'. SD stands for standard deviation, and we have now specified this as follows: 'mean $\pm$ standard deviation (SD)'.

- Table 2: I would recommend removing results that don't survive corrections for multiple comparisons.

We acknowledge the reviewer's concern regarding the reporting of results that do not survive multiple comparisons. However, considering the uniqueness of our dataset and the novelty of our work, we believe it is crucial to report all significant findings. We have taken great care to transparently distinguish between results that survived multiple comparisons and those that did not in both the Results and Discussion sections, ensuring that readers are not misled. It is possible that future studies may replicate and further strengthen these associations. Therefore, by sharing these results with the research community, we provide valuable insights for future investigations.

- Figure 3: the top is redundant with Table 2: to be merged? B: the statistical results might be shown in a Table.

We agree with the reviewer that the top part of Figure 3 and Table 2 report the same results. However, given the richness of these findings, we believe that the top part of Figure 3 serves as a useful summary for readers. Additionally, examining both the top and bottom parts of Figure 3 provides a comprehensive overview of the regression analysis conducted in this study.

- Figure SI6: Is it really a % in x-axis?

We thank the reviewer for spotting this typo, the percentage is relevant for the y-axis only. We removed the % symbol from ticks of the x-axis.

- Table SI1: the presented *p*-values don't seem to survive Bonferroni correction, contrary to what is written.

We thank the reviewer for spotting this mistake, we removed the reference to the Bonferroni correction for the *p*-values.

- Table SI2: For the proportion of children included in the analysis, maybe be precise that the proportion was computed based on the ones with acquired data. Maybe also add the proportion according to all children, to better show the high drop-out rate at certain ages?

We thank the reviewer for these useful suggestions. We have specified in the legend of the table how we calculated the proportion of infants included as follows: 'The proportion of children included in the analysis was computed based on the infants with FC data'. We have also added a column in the table called 'Inclusion rate (from the 204 infants recruited)', following the reviewer's suggestion. This will be a useful reference for future studies.

- A few typos should be corrected throughout the manuscript.

We thoroughly revised the main manuscript and the supplementary materials for typos.

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